

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	19
Flg Calc [Int P]: New Flange	20
Flg Calc [Int P]: LHS Ch Flg	23
Flg Calc [Int P]: Shell Flg	29
Internal Pressure Calculations:	35
Element: Shell Flg	36
Element: Main Shell	38
Element: Dish End	40
Shop Pressure Test Results:	43
Field Pressure Test Results:	45
ASME TS Calc: Pressure Test	47
External Pressure Calculations:	55
Element and Detail Weights:	56
Nozzle Flange MAWP:	58
Wind Load Calculation:	59
Earthquake Load Calculation:	61
Center of Gravity Calculation:	63
Saddle Calcs: Operating Case:	64
Saddle Calcs: Test Case:	79
Saddle Calcs: Ext. Pressure:	92
Nozzle Calcs.: N1-Tube inlet	104
Nozzle Calcs.: Noz N2 Fr30	115
Nozzle Calcs.: N3-Shell Inlet	126
Nozzle Calcs.: N4-Shell outlet	139
ASME TS Calc:	152
Problems/Failures Summary:	162



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-31002-AK.pvdb

Date of Analysis : Jul 4, 2026 1:29pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1100.000 mm ID
 Vessel Design Length, Tangent to Tangent 898.61 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 450 °C
 Channel Side Design Temperature 470 °C
 Shell Side Design Pressure 49.967 kgf/cm²
 Channel Side Design Pressure 163.158 kgf/cm²
 Shop Shell Side Test Pressure 67.554 kgf/cm²
 Field Shell Side Test Pressure 67.554 kgf/cm²
 Shop Channel Side Test Pressure 72.768 kgf/cm²
 Field Channel Side Test Pressure 72.768 kgf/cm²

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-336 F11	2	...	K11572	
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Flange	SA-336 F22	3	...	K21590	
Flange	SA-336 F11	2	...	K11572	
Nozzle	SA-336 F11	2	...	K11572	
Nozzle	SA-335 P11	...	...	K11597	
Re-Pad	SA-387 11	2	...	K11789	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F11	2	...	K11572	
Tubes	SB-167	...	...	N06690	
Tubesheet	SA-336 F22	3	...	K21590	
Flg Bolting	SA-193 B16	...	64 < t <= 100	K14072	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-387 11	2	...	K11789	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Ch Cover	163.163	0.00	166.000	0.0000	N/A	No
LHS Ch Flg	163.161	0.00	165.000	0.0000	N/A	No
Shell Flg	163.161	1.05	181.000	0.0000	N/A	No
Shell Flg	50.077	1.05	52.700	1.5000	N/A	No
Main Shell	50.077	1.05	58.000	1.5000	N/A	No
Dish End	50.077	1.05	54.100	1.5000	No	No

Liquid Level: 166.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
--------------	--------------	-------------------	----------------------	-----------------------	----------------------	----------------------	----------	----------

Body Flg	32.3	32.0	328.0	320.0	311.6	205.6	1.00	1.00
Body Flg	75.8	43.5	318.0	310.0	299.6	183.5	1.00	1.00
Cylinder	255.8	180.0	82.0	82.0	73.1	5.7	1.00	1.00
Body Flg	313.5	29.4	258.0	250.0	246.7	222.1	1.00	1.00
Cylinder	893.5	580.0	24.0	24.0	20.8	9.9	1.00	1.00
Ellipse	898.6	5.1	26.0	22.0	20.4	4.9	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube inlet	WRC-297	0.891	Shell	Passed
Noz N2 Fr30	WRC-297	0.891	Shell	Passed
N3-Shell Inlet	WRC-297	0.859	Shell	Passed
N4-Shell outle	WRC-297	0.857	Shell	Passed

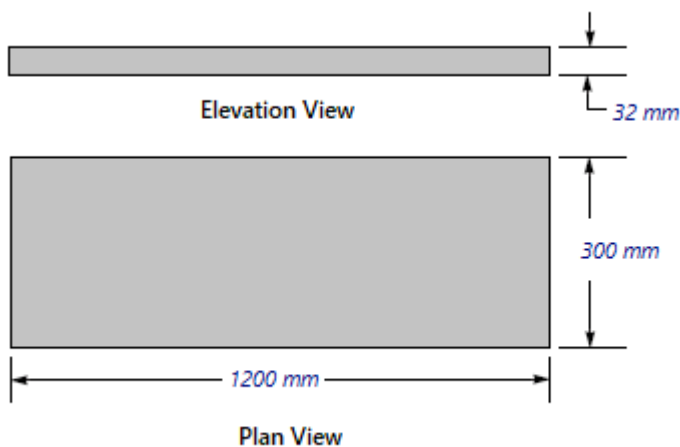
Foundation/Support Summary

Saddle Parameters:

Saddle Width	150.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1150.000 mm
Wear Pad Width	330.000 mm
Wear Pad Thickness	20.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1000.000 mm

Baseplate Length	1200.000 mm
Baseplate Thickness	32.000 mm
Baseplate Width	300.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	20.000 mm
Web Thickness	20.000 mm
Height of Center Web	350.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load 23. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	57905.81 kgf
Maximum Transverse Saddle Shear Load	1961.15 kgf
Maximum Longitudinal Saddle Shear Load	3922.30 kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	10602.50 kgf
------------------------------	--------------

Maximum Transverse Saddle Shear Load	536.02	kgf
Maximum Longitudinal Saddle Shear Load	234.03	kgf

Weights:

Fabricated - Bare w/o removable internals	35408.4	kg
Shop Test - Fabricated + water (full)	42946.5	kg
Shipping - Fab. + rem. intls.+ shipping app.	37383.0	kg
Erected - Fab. + rem. intls.+ insul. (etc)	37383.0	kg
Empty - Fab. + intls. + details + wghts.	37383.0	kg
Operating - Empty + operating liquid (no ca)	42839.2	kg
Field Test - Empty weight + test medium (full)	44661.4	kg

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube inlet	179	OK	...	OK	26.07	OK	Passed
Noz N2 Fr30	173	OK	...	OK	28.81	OK	Passed
N3-Shell Inlet	50.9	OK	...	OK	9.60	OK	Passed
N4-Shell outlet	50.2	OK	...	OK	9.83	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 50.2 Nozzle : N4-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 173 Nozzle : Noz N2 Fr30 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube inlet	1658.175	90.000	739.450
30	Noz N2 Fr30	1658.175	270.000	822.051
60	N3-Shell Inlet	3535.349	270.000	450.400
60	N4-Shell outlet	7935.350	90.000	520.320

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3-Shell Inlet	222 mm	Actual	WNF	273.0	25.400	373.00	24.00	341	900
N4-Shell outlet	257 mm	Actual	WNF	323.8	33.320	423.80	24.00	348	900
N1-Tube inlet	450 mm	160	WNF	457.2	45.237	...	...	402	2500
Noz N2 Fr30	500 mm	160	WNF	508.0	49.987	...	...	419	2500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3-Shell In	SA-335 P11	24.000	10.000	18.000	24.000	...
N4-Shell ou	SA-335 P11	24.000	10.000	18.000	24.000	...
N1-Tube inl	SA-336 F11	24.000	20.000	...	...	...
Noz N2 Fr30	SA-336 F11	24.000	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3-Shell Inlet	353.535	270.0	300.00	0.00	Main Shell
N4-Shell outlet	793.535	90.0	300.00	0.00	Main Shell
N1-Tube inlet	165.817	90.0	200.00	0.00	Shell Flg
Noz N2 Fr30	165.817	270.0	200.00	0.00	Shell Flg

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 328.0mm THK X 1660.0mm OD	SA-336 F22
1	WELD NECK FLANGE: 318.0mm THK X 1660.0mm OD	SA-336 F22
1	CYLINDER: 82.0mm THK X 1100.0mm ID X 180.0cm	SA-336 F11
1	WELD NECK FLANGE: 258.0mm THK X 1560.0mm OD	SA-336 F11
1	CYLINDER: 24.0mm THK X 1100.0mm ID X 580.0cm	SA-387 11
1	ELLIPTICAL HEAD: 2.0 X 1, 26.0mm THK X 1100.0mm ID X 5.1cm	SA-387 11
1	INSULATION: 310mm X 295mm THK	Hot Insulation
1	INSULATION: 339mm X 295mm THK	Hot Insulation
1	INSULATION: 1800mm X 295mm THK	Insulation
9	BAFFLES: 1092.0mm DIA X 10.0mm THK	SA-516 70
8	TIE RODS: 14mm DIA X 4320mm	SA-193 B7M
1	TUBESHEET: 1570.0mm X 280.0mm THK	SA-336 F22
860	TUBES: 5800.0mm X 25.0mm DIA X 2.1mm THK	SB-167
1	INSULATION: 294mm X 295mm THK	Insulation
2	SADDLE: 150.0mm X 120 DEG	SA-387 11
1	INSULATION: 5800mm X 295mm THK	Hot Insulation
1	INSULATION: 325mm X 295mm THK	Hot Insulation
1	GASKET: 1135.0mm OD X 1106.0mm ID	...
48	BODY FLANGE BOLTS: 88.9mm DIA	SA-193 B16
72	NUTS FOR BODY FLANGE BOLTS: 88.9mm DIA	...
1	GASKET: 1135.0mm OD X 1103.0mm ID	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell Flg [11]		B	-2	-11		0.836	24.000	1.00	No
Bolting [21]			-29						
Main Shell [8]		A	18	8		0.832	24.000	1.00	Yes
Left Saddle [24]		A	12	3		0.832	20.000	1.00	Yes
New Saddle [24]		A	12	3		0.832	20.000	1.00	Yes
N3-Shell Inlet[1]		A	18	8		0.830	24.000	1.00	Yes
Nozzle Flg [5]		B	-18	-104					
N4-Shell outle[1]		A	18	8		0.828	24.000	1.00	Yes
Nozzle Flg [5]		B	-18	-104					
Tubesheet: SS[13]		B	24	6		0.665	70.000	1.00	No
Warmest MDMT:			24	8					
LHS Ch Cover [11]		B	27	-104		0.291	80.000	1.00	No
LHS Ch Flg [11]		B	27	-104		0.281	82.000	1.00	Yes
Shell Flg [8]		B	27	-104		0.128	82.000	1.00	Yes
Dish End [10]		A	15	13		0.964	22.000	1.00	Yes
Dish End [7]		A	20	10		0.825	26.000	1.00	Yes
N1-Tube inlet [1]		B	27	18		0.836	82.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
Noz N2 Fr30 [1]		B	27	18		0.836	82.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
Tubesheet: CS[14]		B	27	9		0.665	82.000	1.00	No
Bolting [21]			-29						
Warmest MDMT:			27	18					
Exchanger Side			Computed MDMT		Required MDMT		Pass/Fail		
Shell			8.0		11.4		Pass		
Channel/Tube			18.0		11.4		Fail		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Units used in this Analysis (KgfcM2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	49.967	kgf/cm ²
Channel Side Design Pressure	163.16	kgf/cm ²
Shell Side Design Temperature	450.0	°C
Channel Side Design Temperature	470.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	User Entered Pressure
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM	
Importance Factor for IS-1893	2.000	
Soil Factor	1.000	
Zone Number	2.000	
Percent Seismic for Hydrotest	0.000	
Period of Vibration (User defined) Sec	0.000	
Damping Factor	2.000	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Ch Cover
Distance "FROM" to "TO"	32 cm
Flange Outside Diameter	1660 mm
Element Thickness	320 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	328 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	470 °C
Design External Pressure	0 kgf/cm ²



Design Temperature External Pressure	470	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Allowable Stress, Ambient	1509.1	kgf/cm ²
Allowable Stress, Operating	1211.6	kgf/cm ²
Allowable Stress, Hydrotest	2844.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2308.6	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-3	
UNS Number	K21590	
Class / Thickness / Grade	3 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	166	cm
Liquid Density	24.989	kg/m ³

Element From Node	10	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	31	cm
Thickness of Insulation	295	mm
Density	0.00015	kg/cm ³

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Flange	
Description	LHS Ch Flg	
Distance "FROM" to "TO"	43.5	cm
Flange Inside Diameter	1106	mm
Element Thickness	310	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	318	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	470	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	470	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	110.6	cm
Liquid Density	24.989	kg/m ³

Element From Node	20	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	34	cm
Thickness of Insulation	295	mm
Density	0.00015	kg/cm ³



Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell Flg
Distance "FROM" to "TO"	180 cm
Inside Diameter	1100 mm
Element Thickness	82 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	82 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	450 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	450 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1325.6 kgf/cm ²
Allowable Stress, Hydrotest	2523.7 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	1988.4 kgf/cm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K11572
Class / Thickness / Grade	2 ::
Product Form	Forgings
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	110 cm
Liquid Density	24.989 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	180 cm
Thickness of Insulation	295 mm
Density	0.00015 kg/cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N1-Tube inlet
Dist. from "FROM" Node / Offset dist	90 cm
Nozzle Diameter	450 mm
Nozzle Schedule	160
Nozzle Class	2500
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	549.32 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F11

Element From Node	30
Detail Type	Nozzle
Detail ID	Noz N2 Fr30
Dist. from "FROM" Node / Offset dist	90 cm
Nozzle Diameter	500 mm
Nozzle Schedule	160
Nozzle Class	2500
Layout Angle	270.0
Blind Flange (Y/N)	N



Weight of Nozzle (Used if > 0) 938.14 kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-336 F11

Element# 4/6
Element From Node 40
Element To Node 60
Element Type Flange
Description Shell Flg
Distance "FROM" to "TO" 29.4 cm
Flange Inside Diameter 1100 mm
Element Thickness 250 mm
Internal Corrosion Allowance 1.5 mm
Nominal Thickness 258 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 49.967 kgf/cm²
Design Temperature Internal Pressure 450 °C
Design External Pressure 1.0546 kgf/cm²
Design Temperature External Pressure 450 °C
Effective Diameter Multiplier 1.2
Material Name SA-336 F11
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element From Node 40
Detail Type Liquid
Detail ID Liquid: 40
Dist. from "FROM" Node / Offset dist 0 cm
Height/Length of Liquid 110 cm
Liquid Density 999.54 kg/m³

Element From Node 40
Detail Type Insulation
Detail ID Insulation
Dist. from "FROM" Node / Offset dist 0 cm
Height/Length of Insulation 29.4 cm
Thickness of Insulation 295 mm
Density 0.00015 kg/cm³

Element# 5/6
Element From Node 60
Element To Node 70
Element Type Cylinder
Description Main Shell
Distance "FROM" to "TO" 580 cm
Inside Diameter 1100 mm
Element Thickness 24 mm
Internal Corrosion Allowance 1.5 mm
Nominal Thickness 24 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 49.967 kgf/cm²
Design Temperature Internal Pressure 450 °C
Design External Pressure 1.0546 kgf/cm²
Design Temperature External Pressure 450 °C
Effective Diameter Multiplier 1.2
Material Name SA-387 11
Allowable Stress, Ambient 1509.1 kgf/cm²
Allowable Stress, Operating 1458.2 kgf/cm²
Allowable Stress, Hydrotest 2844.9 kgf/cm²
Material Density 0.00775 kg/cm³
P Number Thickness 16.002 mm
Yield Stress, Operating 2233.1 kgf/cm²
UCS-66 Chart Curve Designation A
External Pressure Chart Name CS-3
UNS Number K11789
Class / Thickness / Grade 2 ::



Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 60
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 100 cm
 Width of Saddle 150 mm
 Height of Saddle at Bottom 1150 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 330 mm
 Thickness of Wear Plate 20 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.3000001
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 60
 Detail Type Saddle
 Detail ID New Saddle
 Dist. from "FROM" Node / Offset dist 480 cm
 Width of Saddle 150 mm
 Height of Saddle at Bottom 1150 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 330 mm
 Thickness of Wear Plate 20 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.0
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 60
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 110 cm
 Liquid Density 999.54 kg/m³

Element From Node 60
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 580 cm
 Thickness of Insulation 295 mm
 Density 0.00015 kg/cm³

Element From Node 60
 Detail Type Nozzle
 Detail ID N3-Shell Inlet
 Dist. from "FROM" Node / Offset dist 40 cm
 Nozzle Diameter 222.2 mm
 Nozzle Schedule 140
 Nozzle Class 900
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 154.27 kgf
 Grade of Attached Flange GR 1.9
 Nozzle Matl SA-335 P11

Element From Node 60



Detail Type	Nozzle
Detail ID	N4-Shell outlet
Dist. from "FROM" Node / Offset dist	480 cm
Nozzle Diameter	257.2 mm
Nozzle Schedule	160
Nozzle Class	900
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	204.65 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-335 P11

Element#	6/6
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Dish End
Distance "FROM" to "TO"	5.08 cm
Inside Diameter	1100 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	49.967 kgf/cm ²
Design Temperature Internal Pressure	450 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	450 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	110 cm
Liquid Density	999.54 kg/m ³

Element From Node	70
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	32.58 cm
Thickness of Insulation	295 mm
Density	0.00015 kg/cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
LHS Ch Cover		32.3175	...	32	...
LHS Ch Flg		75.8175	...	43.5	...
Shell Flg		255.817	...	180	...
Shell Flg		313.535	...	29.4	...
Main Shell		893.535	...	580	...
Dish End		898.615	...	5.08	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

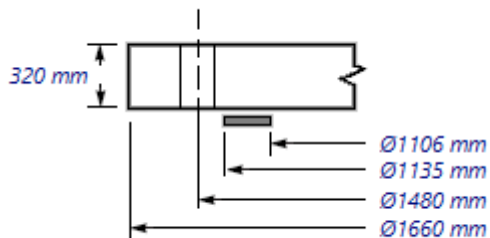
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Flange Input Data Values

Description: New Flange :

LHS Ch Cover

Flange Type	TEMA Channel Cover		
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		470	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1660.000	mm
Flange Thickness	t	320.0000	mm
Flange Material	SA-336 F22		
Flange Material UNS number	K21590		
Flange Allowable Stress At Temperature	Sfo	1211.58	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1509.10	kgf/cm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	1384.92	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1549.89	kgf/cm ²
Perimeter along the Center of the Bolts	L	4649.557	mm
Youngs Modulus for Blind Flange Matl.	Sy	1810721.12	kgf/cm ²
Allowed Channel Cover Deflection	dfAllow	1.3750	mm
Diameter of Bolt Circle	C	1480.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1146.000	mm
Flange Face Inside Diameter	Fid	1106.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1135.000	mm
Gasket Inside Diameter	Gi	1106.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	14.500 mm
Basic Gasket Width,	$bo = N/2$	7.250 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.785 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1121.430 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 1121^2 \cdot 163 \\ &= 1611588.000 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 6.79 \cdot \pi \cdot 1121 \cdot 3 \cdot 163 \\ &= 234017.766 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(1611588 + 234018 + 0, 0) \\ &= 1845605.750 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\ &= 703 \cdot 6.79 \cdot \pi \cdot 1121 + 0 \cdot 0 \cdot 0 \\ &= 168064.359 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((1845605 + 0 + 4 \cdot 0 / 1121) / 1385, 168064 / 1550) \\ &= 1332.642 \text{ cm}^2 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1332.642	1354.681	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	193.179	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= Ab \cdot Sa / (y \cdot \pi (Go + Gi)) \\ &= 1355 \cdot 1550 / (703 \cdot \pi (1135 + 1106)) \\ &= 42.418 \text{ mm [Note: Exceeds actual gasket width, 14.500]} \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned} &= Sa \cdot Ab \\ &= 1550 \cdot 1355 \\ &= 2099599.50 \text{ kgf} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= Wo \\ &= 1845605.75 \text{ kgf} \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned} &= (C - G) / 2 \\ &= (1480 - 1121) / 2 \\ &= 179.2851 \text{ mm} \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned} &= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\ &= 1.9 \cdot 1845605 \cdot 179 \cdot 1 / (320^2 \cdot 1121) + 0.3 \cdot 1 \cdot 163 \cdot 1121^2 / 320^2 \\ &= 1148.63 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned} &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\ &= 1.9 \cdot 2099599 \cdot 179 \cdot 1 / (320^2 \cdot 1121) \\ &= 622.82 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

= 1845605/1355
 = 1362.39 kgf/cm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (168064/1355)
 = 124.06 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1148.63	1211.58	622.82 1509.10
Bolting	1362.39	1384.92	124.06 1549.89

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.5071 mm
 Allowed deflection at center of Cover 1.3750 mm

Reqd. Blind Flange Thickness at Center 311.576 mm
 Reqd. Blind Flange Thickness at Gasket 215.108 mm
 Min. Required Flange Thickness per TEMA 229.474 mm

Estimated M.A.W.P. (Operating) 166 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 5367.6 kg
 Estimated Unfinished Weight of Forging at given Thk 5367.6 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.29, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1148.63	1509.10	622.82 1509.10
Bolting	1362.39	1549.89	124.06 1549.89

Minimum Req'd Thickness, ambient, no Corrosion: 279.197 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Flange Input Data Values

Description: LHS Ch Flg :

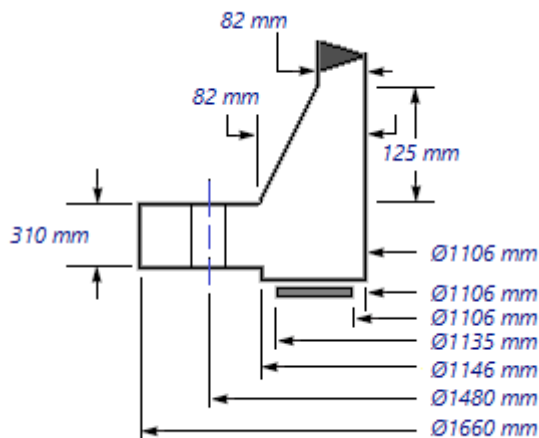
LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		470	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1106.000	mm
Flange Outside Diameter	A	1660.000	mm
Flange Thickness	t	310.0000	mm
Thickness of Hub at Small End	go	82.0000	mm
Thickness of Hub at Large End	gl	82.0000	mm
Length of Hub	h	125.0000	mm
Flange Material	SA-336 F22		
Flange Material UNS number	K21590		
Flange Allowable Stress At Temperature	Sfo	1211.58	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1509.10	kgf/cm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	1384.92	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1549.89	kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1480.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1146.000	mm
Flange Face Inside Diameter	Fid	1106.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1135.000	mm
Gasket Inside Diameter	Gi	1106.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (163 \cdot (1106/2 + 0)) / (1212 \cdot 1 - 0.6 \cdot 163) + Ca$$

$$= 81.0176 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1212 \cdot 1 \cdot 82) / (553 + 0.6 \cdot 82)$$

$$= 164.978 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 82) / (553 + 0.6 \cdot 82)$$

$$= 205.490 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	1106.000 mm
Corroded Large Hub,	g1Cor = g1-ci	82.000 mm
Corroded Small Hub,	g0Cor = go-ci	82.000 mm
Code R Dimension,	R = (C-B)/2 - g1	105.000 mm
Gasket Contact Width,	N = (Go - Gi)/2	14.500 mm
Basic Gasket Width,	bo = N/2	7.250 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.785 mm
Gasket Reaction Diameter,	G = Go - 2 * b	1121.430 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1121^2 \cdot 163$$

$$= 1611574.250 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.79 \cdot \pi \cdot 1121 \cdot 3 \cdot 163$$

$$= 234015.766 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1106^2 \cdot 163$$

$$= 1567532.000 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1611574 - 1567532$$

$$= 44042.309 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1611574 + 234016 + 0, 0)$$

$$= 1845590.125 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 6.79 \cdot \pi \cdot 1121 + 0 \cdot 0$$

$$= 168064.359 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1845590 + 0 + 4 \cdot 0 / 1121) / 1385, 168064 / 1550)$$

$$= 1332.631 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 88.9 + 6 \cdot 310 / (3 + 0.5)$$

$$= 709.229 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 1480 \cdot \sin(\pi/24)$$

$$= 193.179 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{193/(2 \cdot 88.9 + 310)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1332.631	1354.681	
Radial Distance between Hub and Bolts:	104.775	105.000	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	193.179	709.229

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1355 \cdot 1550 / (703 \cdot \pi(1135 + 1106))$$

$$= 42.418 \text{ mm [Note: Exceeds actual gasket width, 14.500]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1550(1333 + 1355) / 2$$

$$= 2082511.62 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1845590 - 1611574$$

$$= 234015.83 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1480 - 1121) / 2$$

$$= 179.2851 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1480 - 1106) / 2 + 179)$$

$$= 183.1426 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1480 - 1106 - 82) / 2$$

$$= 146.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1567532.	146.0000	1.0000	228860.
Face Pressure, Mt	44042.	183.1426	1.0000	8066.
Gasket Load, Mg	234016.	179.2851	1.0000	41956.
Gasket Seating, Matm	2082512.	179.2851	1.0000	373364.

Total Moment for Operation, Mo 278882. kgf-m.
 Total Moment for Gasket seating, Mg 373364. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 301.151 mm
 Hub Ratio, h/h0 = Defined as 2 2.000
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.000



Note: Because $g_1 = g_0$, the ratio h/h_0 has been set to 2.

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.926
Factor V per 4.16.5	0.549
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 1660/1106 \\ &= 1.501 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.5^2 \cdot \log_{10}(1.5))) / (1.5^2 - 1)) / (1.5 - 1) \\ &= 4.954 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.5^2(1 + 8.55246 \cdot \log_{10}(1.5)) - 1) / ((1.0472 + 1.9448 \cdot 1.5^2) \cdot (1.5 - 1)) \\ &= 1.710 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.5^2(1 + 8.55246 \cdot \log_{10}(1.5)) - 1) / (1.36136(1.5^2 - 1) \cdot (1.5 - 1)) \\ &= 5.444 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.5^2 + 1) / (1.5^2 - 1) \\ &= 2.597 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.44 \cdot 82^2 \cdot 301 / 0.55 \\ &= 20074108.000 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.93/301 \\ &= 0.003 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (310 \cdot 0.0031 + 1) / 1.71 + 310^3/20074108 \\ &= 2.626 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 278882/1106) / (2.63 \cdot 82^2) \\ &= 1428.06 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 373364/1106) / (2.63 \cdot 82^2) \\ &= 1911.87 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.27 \cdot 278882/1106) / (2.63 \cdot 310^2) \\ &= 226.89 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.27 \cdot 373364/1106) / (2.63 \cdot 310^2) \\ &= 303.76 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (4.95 \cdot 278882 / (310^2 \cdot 1106)) - 2.6 \cdot 227$$

$$= 710.72 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (4.95 \cdot 373364 / (310^2 \cdot 1106)) - 2.6 \cdot 304$$

$$= 951.51 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1428 + \max(227, 711)) / 2$$

$$= 1069.39 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STa)) / 2$$

$$= (1912 + \max(304, 952)) / 2$$

$$= 1431.69 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1845590 / 1355$$

$$= 1362.38 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (168064 / 1355)$$

$$= 124.06 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1428.06	1817.37	1911.87	2263.65
Radial Flange	226.89	1211.58	303.76	1509.10
Tangential Flange	710.72	1211.58	951.51	1509.10
Maximum Average	1069.39	1211.58	1431.69	1509.10
Bolting	1362.38	1384.92	124.06	1549.89

Minimum Required Flange Thickness 299.644 mm

Estimated M.A.W.P. (Operating) 165 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 3188.1 kg
 Estimated Unfinished Weight of Forging at given Thk 4057.5 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 373364 / 1 \cdot 100000 \cdot 0.55 / (2.46 \cdot 2151394 \cdot 82^2 \cdot 301 \cdot 0.3)$$

$$= 0.332 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 278882 / 1 \cdot 100000 \cdot 0.55 / (2.46 \cdot 1810721 \cdot 82^2 \cdot 301 \cdot 0.3)$$

$$= 0.294 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 373364 / 1 \cdot 100000 \cdot 0.55 / (2.63 \cdot 2151394 \cdot 82^2 \cdot 301 \cdot 0.3)$$

$$= 0.311 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 278882 / 1 \cdot 100000 \cdot 0.55 / (2.63 \cdot 1810721 \cdot 82^2 \cdot 301 \cdot 0.3)$$

$$= 0.276 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Thickness Ratio = 0.28, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 27 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1428.06	2263.65	1810.96	2263.65
Radial Flange	226.89	1509.10	287.72	1509.10
Tangential Flange	710.72	1509.10	901.29	1509.10
Maximum Average	1069.39	1509.10	1356.12	1509.10
Bolting	1362.38	1549.89	124.06	1549.89

Minimum Req'd Thickness, ambient, no Corrosion: 288.950 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Flange Input Data Values

Description: Shell Flg :

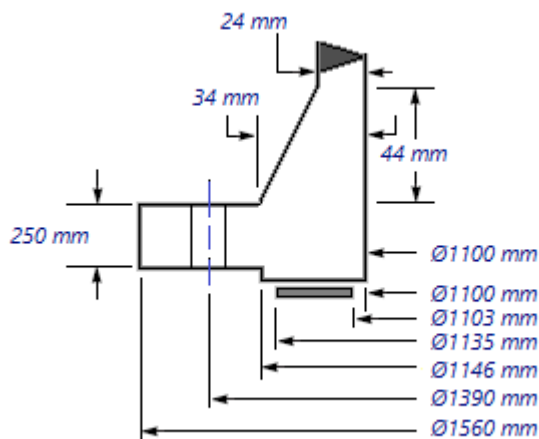
Shell Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	50.08	kgf/cm ²
Design Temperature		450	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1560.000	mm
Flange Thickness	t	250.0000	mm
Thickness of Hub at Small End	go	24.0000	mm
Thickness of Hub at Large End	gl	34.0000	mm
Length of Hub	h	44.0000	mm
Flange Material		SA-336 F11	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	1325.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B16	
Bolt Allowable Stress At Temperature	Sb	1384.92	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1549.89	kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1390.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1146.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1135.000	mm
Gasket Inside Diameter	Gi	1103.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	175.77	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (50.1 \cdot (1100/2 + 1.5)) / (1326 \cdot 1 - 0.6 \cdot 50.1) + Ca$$

$$= 22.8171 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1326 \cdot 1 \cdot 22.5) / (552 + 0.6 \cdot 22.5)$$

$$= 52.790 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 24) / (550 + 0.6 \cdot 24)$$

$$= 59.836 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	248.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1103.000 mm
Corroded Large Hub, $g_{1cor} = g_1 - c_i$	32.500 mm
Corroded Small Hub, $g_{0cor} = g_0 - c_i$	22.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g_{1cor}$	111.000 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $b_o = N/2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1120.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1121^2 \cdot 50.1$$

$$= 494018.500 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1121 \cdot 2 \cdot 50.1$$

$$= 50267.582 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1103^2 \cdot 50.1$$

$$= 478498.344 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 494018 - 478498$$

$$= 15520.161 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(494018 + 50268 + 0, 0)$$

$$= 544286.062 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 176 \cdot 7.13 \cdot \pi \cdot 1121 + 0 \cdot 0 \cdot 0$$

$$= 44108.797 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((544286 + 0 + 4 \cdot 0 / 1121) / 1385, 44109 / 1550)$$

$$= 393.008 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 88.9 + 6 \cdot 248 / (2 + 0.5)$$

$$= 774.200 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1390 \cdot \sin(\pi / 24)$$

$$= 181.431 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{181 / (2 \cdot 88.9 + 248)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	393.008	1354.681	
Radial Distance between Hub and Bolts:	104.775	111.000	
Radial Distance between Bolts and Edge:	82.550	85.000	
Circ. Spacing between the Bolts:	180.975	181.431	774.200

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1355 \cdot 1550 / (176 \cdot \pi(1135 + 1103))$$

$$= 169.899 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1550(393 + 1355) / 2$$

$$= 1354358.62 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 544286 - 494018$$

$$= 50267.57 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1390 - 1121) / 2$$

$$= 134.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1390 - 1103) / 2 + 135)$$

$$= 139.0637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1390 - 1103 - 32.5) / 2$$

$$= 127.2500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	478498.	127.2500	1.0000	60889.
Face Pressure, Mt	15520.	139.0637	1.0000	2158.
Gasket Load, Mg	50268.	134.6274	1.0000	6767.
Gasket Seating, Matm	1354359.	134.6274	1.0000	182334.

Total Moment for Operation, Mo 69815. kgf-m.
 Total Moment for Gasket seating, Mg 182334. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 157.536 mm
 Hub Ratio, h/h0 = HL / H0 0.279



Thickness Ratio, $g_1/g_0 = (g_1Cor/goCor)$ 1.444

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.888
 Factor V per 4.16.5 0.389
 Factor f per 4.16.5 1.148

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1560/1103
 = 1.414

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(1.41^2 \cdot \log_{10}(1.41))) / (1.41^2 - 1)) / (1.41 - 1)$
 = 5.767

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(1.41^2(1 + 8.55246 \cdot \log_{10}(1.41)) - 1) / ((1.0472 + 1.9448 \cdot 1.41^2) \cdot (1.41 - 1))$
 = 1.748

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(1.41^2(1 + 8.55246 \cdot \log_{10}(1.41)) - 1) / (1.36136(1.41^2 - 1) \cdot (1.41 - 1))$
 = 6.338

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(1.41^2 + 1) / (1.41^2 - 1)$
 = 2.999

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
 = $6.34 \cdot 22.5^2 \cdot 158 / 0.39$
 = 1299420.625 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
 = 0.89/158
 = 0.006 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(248 \cdot 0.0056 + 1) / 1.75 + 248^3/1299420$
 = 13.182

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
 = $(1.15 \cdot 69815 / 1103) / (13.2 \cdot 32.5^2)$
 = 521.71 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
 = $(1.15 \cdot 182334 / 1103) / (13.2 \cdot 32.5^2)$
 = 1362.53 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(2.87 \cdot 69815 / 1103) / (13.2 \cdot 248^2)$
 = 22.29 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(2.87 \cdot 182334 / 1103) / (13.2 \cdot 248^2)$
 = 58.21 kgf/cm²

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (5.77 \cdot 69815 / (248^2 \cdot 1103)) - 3 \cdot 22.3$$

$$= 524.29 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (5.77 \cdot 182334 / (248^2 \cdot 1103)) - 3 \cdot 58.2$$

$$= 1369.27 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (522 + \max(22.3, 524)) / 2$$

$$= 523.00 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STa)) / 2$$

$$= (1363 + \max(58.2, 1369)) / 2$$

$$= 1365.90 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 544286 / 1355$$

$$= 401.78 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (44109 / 1355)$$

$$= 32.56 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	521.71	1988.41	1362.53	2110.70
Radial Flange	22.29	1325.61	58.21	1407.13
Tangential Flange	524.29	1325.61	1369.27	1407.13
Maximum Average	523.00	1325.61	1365.90	1407.13
Bolting	401.78	1384.92	32.56	1549.89

Minimum Required Flange Thickness 246.736 mm

Estimated M.A.W.P. (Operating) 52.8 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 1897.1 kg
 Estimated Unfinished Weight of Forging at given Thk 2189.8 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 182334 / 1 \cdot 100000 \cdot 0.39 / (12.7 \cdot 2081087 \cdot 22.5^2 \cdot 158 \cdot 0.3)$$

$$= 0.584 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 69815 / 1 \cdot 100000 \cdot 0.39 / (12.7 \cdot 1776273 \cdot 22.5^2 \cdot 158 \cdot 0.3)$$

$$= 0.262 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 182334 / 1 \cdot 100000 \cdot 0.39 / (13.2 \cdot 2081087 \cdot 22.5^2 \cdot 158 \cdot 0.3)$$

$$= 0.563 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 69815 / 1 \cdot 100000 \cdot 0.39 / (13.2 \cdot 1776273 \cdot 22.5^2 \cdot 158 \cdot 0.3)$$

$$= 0.252 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Thickness Ratio = 0.84, Temperature Reduction per Fig. UCS 66.1 = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -11 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	523.16	2110.70	1326.00	2110.70
Radial Flange	24.22	1407.13	61.39	1407.13
Tangential Flange	512.48	1407.13	1298.93	1407.13
Maximum Average	517.82	1407.13	1312.47	1407.13
Bolting	401.78	1549.89	32.56	1549.89

Minimum Req'd Thickness, ambient, no Corrosion: 242.265 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
LHS Ch Cover		163.16	328	...	1660	1211.6	1
LHS Ch Flg		163.16	318	...	1106	1211.6	2
Shell Flg		163.16	82	...	1100	1325.6	3
Shell Flg		50.077	258	1.5	1100	1325.6	4
Main Shell		50.077	24	1.5	1100	1458.2	5
Dish End		50.077	26	1.5	1100	1458.2	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
LHS Ch Cover		163.158	166	186	320	311.576	1
LHS Ch Flg		163.158	165	186	310	299.644	2
Shell Flg		163.158	181	193	82	73.0943	3
Shell Flg		49.9673	52.7	59.8	250	246.736	4
Main Shell		49.9673	58	64.2	24	20.8381	5
Dish End		49.9673	54.1	60.1	22	20.4363	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

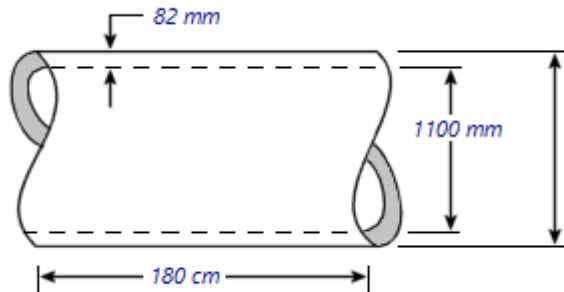
Shell Side MAWP = 50.2 kgf/cm²
 Shell Side MAPnc = 59.8 kgf/cm²
 Channel Side MAWP = 54.1 kgf/cm²
 Channel Side MAPnc = 60.1 kgf/cm²

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell Flg

Cylindrical Shell Design Information:				
Design Pressure	P	163.161 kgf/cm ²	Int. Design Temperature	450 °C
Material		SA-336 F11	Ext. Design Temperature	450 °C
Length		180.000 cm	External Pressure Chart	CS-2
			UNS Number	K11572
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S 1325.6 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa 1407.1 kgf/cm ²
Inside Diameter	Di	1100.000 mm	Actual Stress	1192.3 kgf/cm ²
Spec. Min. Thk t _e , or s _a		82.0000 mm	Spec. Nominal Thk.	82.000 mm
Surface Area		71477.5 cm ²	SG of Contents	0.025
Element Volume		1710597.2 cm ³	Weight of Contents	42.7 kgf
Empty Weight		4247.9 kgf	Operating Weight	4290.7 kgf
			Comp. All. Ope. Stress	933.9 kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (163 \cdot 550) / (1326 \cdot 1 - 0.6 \cdot 163)$$

$$= 73.0943 + 0.0000 = 73.0943 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.003 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1326 \cdot 1 \cdot 82) / (550 + 0.6 \cdot 82)$$

$$= 181 - 0.0027 = 181 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 82) / (550 + 0.6 \cdot 82)$$

$$= 193 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (163 \cdot (550 + 0.6 \cdot 82)) / (1 \cdot 82)$$

$$= 1192.270 \text{ kgf/cm}^2$$

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 82, tr = 10.5, c = 0 mm, E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	54.087 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	82.000 mm	Basic MDMT	27 °C
Stress/Thk. Ratio	0.128	Temperature Difference	131 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell Flg

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 450 °C

Elastic Modulus from Chart: CS-2 at 450 °C: 1544036 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
82.000	1264.00	2235.00	15.41	1.7682	0.0121409	750.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*750)/(3*15.4) = 64.9 \text{ kgf/cm}^2$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.733	1264.00	2235.00	220.46	1.7682	0.0002259	174.39

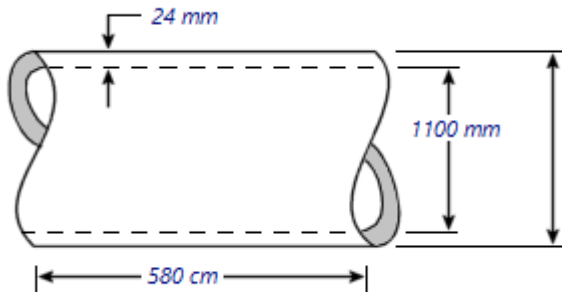
$$MAEP = (4*B)/(3*(Do/t)) = (4*174)/(3*220) = 1.05 \text{ kgf/cm}^2$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
82.000	1264.00	128337.89	15.41	50.0000	0.0046803	651.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*652)/(3*15.4) = 56.4 \text{ kgf/cm}^2$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	50.077 kgf/cm²	Int. Design Temperature	450 °C	
Material		SA-387 11	Ext. Design Temperature	450 °C	
Length		580.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1458.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	1100.000 mm	Actual Stress	1257.5 kgf/cm²	
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		209179.8 cm²	SG of Contents	1.000	
Element Volume		2960699.0 cm³	Weight of Contents	2959.4 kgf	
Empty Weight		3809.6 kgf	Operating Weight	6769.0 kgf	
			Comp. All. Ope. Stress	788.3 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (50.1 \cdot 552) / (1458 \cdot 1 - 0.6 \cdot 50.1)$$

$$= 19.3381 + 1.5000 = 20.8381 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.110 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1458 \cdot 1 \cdot 22.5) / (552 + 0.6 \cdot 22.5)$$

$$= 58.1 - 0.11 = 58 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1509 \cdot 1 \cdot 24) / (550 + 0.6 \cdot 24)$$

$$= 64.2 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (50.1 \cdot (552 + 0.6 \cdot 22.5)) / (1 \cdot 22.5)$$

$$= 1257.498 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.135 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24, tr = 18.7, c = 1.5 mm, E* = 1

Thickness Ratio = $tr * E / (tg - c) = 0.83$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	50.212 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm	Basic MDMT	18 °C
Stress/Thk. Ratio	0.832	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	8 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 450 °C

Elastic Modulus from Chart: CS-2 at 450 °C: 1544036 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
22.500	1148.00	6236.47	51.02	5.4325	0.0005859	399.01

$$MAEP = (4*B)/(3*(Do/t)) = (4*399)/(3*51) = 10.4 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.426	1148.00	6236.47	136.25	5.4325	0.0001396	107.78

$$MAEP = (4*B)/(3*(Do/t)) = (4*108)/(3*136) = 1.05 \text{ kgf/cm}^2$$

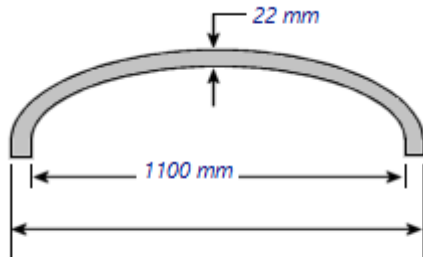
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
22.500	1148.00	159916.89	51.02	50.0000	0.0004245	327.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*328)/(3*51) = 8.56 \text{ kgf/cm}^2$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	50.077 kgf/cm²	Int. Design Temperature	450 °C	
Material		SA-387 11	Ext. Design Temperature	450 °C	
Length		5.080 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1458.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	1100.000 mm	Actual Stress	1347.3 kgf/cm²	
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.	26.000 mm	
Surface Area		16224.1 cm²	SG of Contents	1.000	
Element Volume		222504.4 cm³	Weight of Contents	222.4 kgf	
Empty Weight		342.0 kgf	Operating Weight	564.4 kgf	
			Comp. All. Ope. Stress	788.3 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (50.1 \cdot 1103 \cdot 1) / (2 \cdot 1458 \cdot 1 - 0.2 \cdot 50.1)$$

$$= 18.9363 + 1.5000 = 20.4363 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.110 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1458 \cdot 1 \cdot 20.5) / (1 \cdot 1103 + 0.2 \cdot 20.5)$$

$$= 54.2 - 0.11 = 54.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1509 \cdot 1 \cdot 22) / (1 \cdot 1100 + 0.2 \cdot 22)$$

$$= 60.1 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.1 \cdot (1 \cdot 1103 + 0.2 \cdot 20.5)) / (2 \cdot 1 \cdot 20.5)$$

$$= 1347.345 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (50.1 \cdot 552) / (1458 \cdot 1 - 0.6 \cdot 50.1) + 1.5$
 $= 20.838 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.110 kgf/cm²
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1458 \cdot 1 \cdot 24.5) / (552 + 0.6 \cdot 24.5)$
 $= 63.1 - 0.11 = 63 \text{ kgf/cm}^2$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1103 / (2 \cdot 276))^2) / 6$
 $= 0.996388$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.750 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 22, tr = 19.8, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 15 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 13 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 26, tr = 20.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 10 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	54.087 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	26.000 mm	Basic MDMT	20 °C
Stress/Thk. Ratio	0.825	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	10 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

EXTERNAL PRESSURE RESULTS for: Dish End

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 450 °C

Elastic Modulus from Chart: CS-2 at 450 °C: 1544036 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
20.500	1144.00	55.80	0.0024888	580.55

MAEP = $B / (K_0 \cdot Do / t) = 581 / (0.9 \cdot 55.8) = 11.6 \text{ kgf/cm}^2$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
3.404	1144.00	336.10	0.0004132	319.03

$$MAEP = B / (K0 * Do / t) = 319 / (0.9 * 336) = 1.05 \text{ kgf/cm}^2$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	67.554 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	67.225 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	77.620 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	57.161 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	71.453 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	71.453 kgf/cm ²
App 27-4	= M.A.W.P.	50.212 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	72.768 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	67.225 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	77.995 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	61.573 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	71.453 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	71.453 kgf/cm ²
App 27-4	= M.A.W.P.	54.087 kgf/cm ²

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Shell Flg	532.5	2523.7	0.211	72.88
Main Shell	1699.1	2844.9	0.597	67.66
Dish End	1960.8	2844.9	0.689	72.88

Theoretical Maximum Allowable test pressure Shellside is : 113.181 kgf/cm²

Theoretical Maximum Allowable test pressure Tubeside : 105.626 kgf/cm²

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	1407.13	1325.61	1.062
Noz N2 Fr30	Nozzle	1407.13	1325.61	1.062
N3-Shell Inlet	Nozzle	1203.20	988.09	1.218
N3-Shell Inlet	Pad	1509.10	1458.19	1.035
N4-Shell outlet	Nozzle	1203.20	988.09	1.218
N4-Shell outlet	Pad	1509.10	1458.19	1.035

Minimum 1.035

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Ch Cover	1509.10	1211.58	1.246
LHS Ch Flg	1509.10	1211.58	1.246
Shell Flg	1407.13	1325.61	1.062
Shell Flg	1407.13	1325.61	1.062
Main Shell	1509.10	1458.19	1.035
Dish End	1509.10	1458.19	1.035

Minimum 1.035

Stress ratios for Exchanger Materials (kgf/cm²):



Description	Ambient	Operating	Ratio
Tube Material	1641.66	1580.49	1.039
Tubesheet Material	1509.10	1211.58	1.246

Minimum 1.039

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	351.76	2523.66	0.139
Noz N2 Fr30	352.62	2523.66	0.140
N3-Shell Inlet	359.39	1564.16	0.230
N4-Shell outlet	317.21	1564.16	0.203

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	67.554 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	67.225 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	77.620 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	57.161 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	71.453 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	71.453 kgf/cm ²
App 27-4	= M.A.W.P.	50.212 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	72.768 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	67.225 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	77.995 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	61.573 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	71.453 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	71.453 kgf/cm ²
App 27-4	= M.A.W.P.	54.087 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Shell Flg	532.5	2523.7	0.211	72.88
Main Shell	1699.1	2844.9	0.597	67.66
Dish End	1960.8	2844.9	0.689	72.88

Theoretical Maximum Allowable test pressure Shellside is : 113.181 kgf/cm²

Theoretical Maximum Allowable test pressure Tubeside : 105.626 kgf/cm²

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	1407.13	1325.61	1.062
Noz N2 Fr30	Nozzle	1407.13	1325.61	1.062
N3-Shell Inlet	Nozzle	1203.20	988.09	1.218
N3-Shell Inlet	Pad	1509.10	1458.19	1.035
N4-Shell outlet	Nozzle	1203.20	988.09	1.218
N4-Shell outlet	Pad	1509.10	1458.19	1.035

Minimum 1.035

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Ch Cover	1509.10	1211.58	1.246
LHS Ch Flg	1509.10	1211.58	1.246
Shell Flg	1407.13	1325.61	1.062
Shell Flg	1407.13	1325.61	1.062
Main Shell	1509.10	1458.19	1.035
Dish End	1509.10	1458.19	1.035

Minimum 1.035

Stress ratios for Exchanger Materials (kgf/cm²):



Description	Ambient	Operating	Ratio
Tube Material	1641.66	1580.49	1.039
Tubesheet Material	1509.10	1211.58	1.246

Minimum 1.039

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	351.76	2523.66	0.139
Noz N2 Fr30	352.62	2523.66	0.140
N3-Shell Inlet	359.39	1564.16	0.230
N4-Shell outlet	317.21	1564.16	0.203

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		67.55 kgf/cm ²
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1100.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	1509.10 kgf/cm ²
Shell Allowable Stress at Ambient		1509.10 kgf/cm ²

Channel Data:

Channel Description: Shell Flg

Channel Type:		Cylinder
Channel Test Pressure		72.77 kgf/cm ²
Channel Thickness	tc	82.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1100.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11572
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	860
Tube Wall Thickness	et	2.1000 mm
Tube Outside Diameter	D	25.0000 mm
Total Straight Tube Length	Lt	5800.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5520.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SB-167
Tube Material UNS Number		N06690
Is this a Welded Tube		No
Tube Material Specification used	Smls. pipe & tube	
Tube Allowable Stress at Temperature		1641.66 kgf/cm ²
Tube Allowable Stress At Ambient		1641.66 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2447.19 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	32.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1000 mm
Groove Weld Leg	ag	2.1000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	520.365 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-336 F22



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	1509.10 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1509.10 kgf/cm ²
Thickness of Tubesheet	h	280.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1570.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	1.0546 kgf/cm ²

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	1570.000 mm
Flange Inside Diameter	B	1100.000 mm
Flange Face Outside Diameter	Fod	1146.000 mm
Flange Face Inside Diameter	Fid	1100.000 mm
Gasket Outside Diameter	Go	1135.000 mm
Gasket Inside Diameter	Gi	1103.000 mm
Small end Hub thk.	g0	24.0000 mm
Large end Hub thk.	g1	34.0000 mm
Gasket Factor,	m	2.00
Gasket Design Seating Stress	y	175.77 kgf/cm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	1390.000 mm
Nominal Bolt Diameter	dB	88.9000 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	24
Bolt Material		SA-193 B16
Bolt Allowable Stress At Temperature	Sb	1384.92 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1549.89 kgf/cm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with		Channel
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	248.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes



Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.4870	25.00	6040.37	16.81	50.0000	0.0038914	851.95
$EMAWP = (4*B)/(3*(D/T)) = (4 * 852)/(3 * 16.8) = 67.6 \text{ kgf/cm}^2$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4870 mm
Allowable Internal Pressure at Corroded thickness	295.67 kgf/cm ²
Required Internal Design Pressure	73.82 kgf/cm ²
Allowable External Pressure at Corroded thickness	102.15 kgf/cm ²
Required External Design Pressure	67.55 kgf/cm ²
Required Thickness due to Shell Side pressure	1.4870 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	16.000 mm
Basic Gasket Width,	$b_0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	1120.750 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 520 + 25 \\ &= 1065.730 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (32 - 25) / 32 \\ &= 0.219 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= G_s / D_o \text{ (Configurations d, e, f)} \\ &= 1121 / 1066 \\ &= 1.0516 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1100 / 1066 \\ &= 1.0322 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 280 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2t_t(E_t/E) \cdot (S_t/S) \cdot (p); dt - 2t_t) \\ &= \max(25 - 2 \cdot 2.1 (2130302/2151394) \cdot \\ &\quad (1642/1509) \cdot (0); 25 - 2 \cdot 2.1) \cdot \\ &= 25.0000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$



$$= 32 / ((1 - 4 \cdot \min(0; 4 \cdot 1066 \cdot 32) / (\pi \cdot 1066^2)))$$

$$= 32.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32 - 25) / 32$$

$$= 0.2187$$

$$h/p = 8.750001; \mu^* = 0.218750; E^* = 382522 \text{ kgf/cm}^2$$

$$E^*/E = 0.177802; \nu^* = 0.432718$$

Note: As h/p (8.750) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1100 + 82) 82)$$

$$= 0.0058 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0058 \cdot 2081087 \cdot 82^3 / (6(1 - 0.3^2))$$

$$= 1227097984.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1100 + 82 / \cos(0)) \cdot 1227097984 / 280^3 \cdot (1 + 280 \cdot 0.0058 \cdot (1 + 1) / 2 + (280 \cdot 0.0058)^2 / 2 \cdot 1)$$

$$= 1574415.0 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1100^2 / (4 \cdot 2081087 \cdot 82) \cdot (1 - 1100 \cdot 0.3 / (2(1100 + 82)))$$

$$= 0.15252\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 1227097984 [0.001525193 (1 + 0.0058 \cdot 1 \cdot 280) - 0.000 (1/280 + 1 \cdot 0.0058/2)]$$

$$= 29720.6172 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1570 / 1066$$

$$= 1.4732$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 382522 \cdot (1574415 + 2151394 \cdot \ln(1.47))$$

$$= 3.5709$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1066^2 / 16 \cdot [(1.05 - 1) \cdot (1.05^2 + 1) \cdot 67.6 -$$



$$(1.03 - 1) \cdot (1.03^2 + 1) \cdot 72.8] \\ = 178306.1562 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]
= $M_{ts} + (\omega_c \cdot P_t) + W \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$
= $178306 + (2162712) + 2099599 \cdot (1390 - 1121) / (2 \cdot \pi \cdot 1066)$
= $10783403.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
= $(M_o - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$
= $((10783403) - 1066^2/32 \cdot 3.57(67.6 - 72.8)) / (1 + 3.57)$
= $2503697.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
= $M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$
= $2503697 + 1066^2/64 \cdot (3 + 0.43) \cdot (67.6 - 72.8)$
= $2186086.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
= $\max(|M_p|; |M_o|)$
= $\max(|2503697|; |2186086|)$
= $2503697.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
= $6 \cdot M / (\mu \cdot (h - h_g')^2)$
= $6 \cdot 2503697 / (0.22(280 - 8)^2)$
= $928.2118 \text{ kgf/cm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
= $2 \cdot \beta T \cdot S_y$
= $2 \cdot 0.95 \cdot 3161$
= 6005.81 kgf/cm^2

Tubesheet bending stress at final thickness [σ]:
= $6 \cdot M / (\mu \cdot (h - h_g')^2)$
= $6 \cdot 224489 / (0.22(40 - 8)^2)$
= $6000.7764 \text{ kgf/cm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
= $h + C_{ats} + C_{atc}$
= $40 + 0 + 0$
= 40.0329 mm

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
= $\max(H_{reqB}; H_{reqS})$
= $\max(40; 2.64)$
= 40.0329 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:
= $|P_s - P_t|$
= $|67.6 - 72.8|$
= 5.214 kgf/cm^2

Shear Stress check [τ_{limit}]:
= $4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$
= $4 \cdot 0.22 \cdot 280 / 1066 \cdot 0.8 \cdot 0.95 \cdot 3161$
= 552.269 kgf/cm^2

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:
= $1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc}$
= $1 / (4 \cdot 0.22) \cdot (1066 / 2402) \cdot |67.6 - 72.8| + 0$



= 2.6433 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 82.000 mm
 Min. Channel len. adjacent to the tubesheet is: 540.600mm

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25 - 2.1) 2.1$
 $= 1.511 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.1 (25 - 2.1) 2325$
 $= 3512.330 \text{ kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
 $= 0.85 \cdot \pi \cdot 2.1 (25 + 0.67 \cdot 2.1) 2325$
 $= 3442.6897 \text{ kgf}$

Weld Strength Factor [fw]:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 2325/2325$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 2.1 (25 + 0.67 \cdot 2.1) 2325$
 $= 2227.6228 \text{ kgf}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 25)^2 + 1.07 \cdot 2.1 \cdot (25 - 2.1) \cdot 1 \cdot 1)} - 0.75 \cdot 25 \right)$
 $= 2.651 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.3253 mm
 Minimum Required Groove Weld Leg [agr]: 1.3253 mm

Tube-Tubesheet Jt allowable, 3512 is >= tube strength 3512 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	928.2 <=	6005.8 kgf/cm ²	Ok
Tubesheet shear stress	22.7 <=	2402.3 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	40.033	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.325	2.100	Ok
Tube-Tubesheet Groove Weld Leg :	1.325	2.100	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:



Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	115.548	126.323	906	6006	321	2402	Ps+Pt D1	Ok
D2uc	110.349	...	1558	6006	294	2402	Ps+Pt D2	Ok
D3uc	40.033	...	928	6006	23	2402	Ps+Pt D3	Ok
Max:	115.5475	126.323	mm	0.259		0.134	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
Case#	Shell Channel					
D1uc	...	...	...	...	3996	Ok
D2uc	...	...	...	...	3996	Ok
D3uc	...	...	...	...	3996	Ok
Max	...	...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	115.548	280.000	Ok
Tubesheet thickness flanged extension :	126.323	248.000	Ok
Tube thickness :	1.487	2.100	Ok
Tube-tubesheet fillet weld leg :	1.325	2.100	Ok
Tube-tubesheet groove weld leg :	1.325	2.100	Ok

Given Shell Thickness: 24.0000 mm
 Given Channel Thickness: 82.0000 mm

Min Channel length of thk, (82.000) adj. to tubesheet: 540.600 mm

Note: This is a full strength Tube to Tubesheet Joint.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	320	No Calc	No Calc
20	30	No Calc	...	310	No Calc	No Calc
30	40	223.5	1264	82	0.012141	750.476
40	60	No Calc	...	248.5	No Calc	No Calc
60	70	623.647	1148	22.5	0.00058594	399.01
70	80	No Calc	1144	20.5	0.0024888	580.555

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	320	205.576	...	No Calc
20	30	310	183.54	...	No Calc
30	40	82	5.73339	1.05461	64.9
40	60	250	222.148	1.05461	No Calc
60	70	24	9.92556	1.05461	10.4
70	80	22	4.90378	1.05461	11.6

Minimum

10.4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	223.5	12833.8	No Calc	No Calc
40	60	No Calc	No Calc	No Calc	No Calc
60	70	623.647	15991.7	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	5501.77	...	5501.77	...	550.177
20	30	3262.7	425602	3262.7	425602	326.27
30	40	4247.92	1710597	4247.92	1710597	424.792
40	60	1956.71	288339	1944.56	288568	195.671
60	70	3809.58	2960699	3576.25	2990805	380.958
70	80	341.994	222504	322.264	224197	34.1994
Total		19120	5607742.50	18855	5639770.00	1912



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
10	Liqd	...	16	...	...	Liquid: 10
10	Insl	141.68	15.5	...	...	Hot Insulation
20	Liqd	10.6353	21.75	...	...	Liquid: 20
20	Insl	94.4501	17	...	...	Hot Insulation
30	Liqd	42.7457	90	...	...	Liquid: 30
30	Insl	390.555	90	...	...	Insulation
30	Nozl	604.253	90	73.3363	...	N1-Tube inlet
30	Nozl	1031.95	90	-75.4013	...	Noz N2 Fr30
40	Liqd	288.21	14.7	...	...	Liquid: 40
40	Insl	69.0791	14.7	...	...	Insulation
60	Sadl	378.432	100	82.8	...	Left Saddle
60	Sadl	378.432	480	82.8	...	New Saddle
60	Liqd	2959.37	290	...	...	Liquid: 60
60	Insl	1164.82	290	...	...	Hot Insulation
60	Nozl	169.701	40	-66.11	...	N3-Shell Inlet
60	Nozl	225.118	480	67.858	...	N4-Shell outlet
70	Liqd	222.404	14.2467	...	...	Liquid: 70
70	Insl	113.961	16.29	...	...	Hot Insulation
30	FTsh	3613.56	194	...	...	
30	Tube	7521.16	470	...	...	Tubes
40	Bafl	50.295	74	...	...	Baffles
40	Bafl	50.295	132.8	...	...	Baffles
40	Bafl	50.295	191.6	...	...	Baffles
40	Bafl	50.295	250.4	...	...	Baffles
40	Bafl	50.295	309.2	...	...	Baffles
40	Bafl	50.295	368	...	...	Baffles
40	Bafl	50.295	426.8	...	...	Baffles
40	Bafl	50.295	485.6	...	...	Baffles
40	Bafl	50.7195	544.4	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	756.9
Liquid	3523.4
Insulation	1974.5
Nozzles	2031.0
Exchanger Components	11587.8
Liquid in Tubes	1932.9

Sum of the Detail Weights 21806.5 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	21032.7	21032.7	21032.7	21032.7	21032.7	21032.7
Saddles	756.9	756.9	756.9	756.9	756.9	756.9
Nozzles	2031.0	2031.0	2031.0	2031.0	2031.0	2031.0
Exchanger	11587.8	11587.8	11587.8	11587.8	11587.8	11587.8
Insulation	...	...	1974.5	1974.5	1974.5	1974.5
Ope. Liquid	...	...	...	...	...	3523.4
Tube Ope Lqd	...	...	...	...	...	1932.9
Test Liquid	...	5605.2	...	...	...	...
Tube Tst Lqd	...	1932.9	...	...	...	...
Totals	35408.4	42946.5	37383.0	37383.0	37383.0	42839.2

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	35408.4 kg
Shop Test Wt.	- Fabricated weight + water (full)	42946.5 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	37383.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	37383.0 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	37383.0 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	42839.2 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	42547.5 kg
Field Test Wt.	- Empty weight + test medium (full)	44661.4 kg

Exchanger Tube Data

Volume of Exchanger tubes :	1933755.8 cm ³
Weight of Ope Liq in tubes :	1932.9 kg
Weight of Water in tubes :	1932.9 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	38747.8	...
20	30	33606.3	15392.5
30	40	71477.5	62203.5
40	60	23868.9	10436.4
60	70	209180	200434
70	80	16224.1	14871.7

Total	393104.5	303337.8 cm ²
Total	423.1	326.5 ft ²

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-Tube inlet	195.20	433.49	450	2500	GR 1.1	...	...	...	...
Noz N2 Fr30	195.20	433.49	450	2500	GR 1.1	...	...	...	...
N3-Shell Inlet	108.60	158.16	450	900	GR 1.9	60.571	02.334	08.604	02.334
N4-Shell outlet	108.60	158.16	450	900	GR 1.9	47.603	93.582	08.604	08.604

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 93.582 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 158.162 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 195.198 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 433.491 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N3-Shell Inlet	239.776	0.50	1588	857	50.212	102.334	Pass
N4-Shell outlet	293.751	0.30	2294	1030	50.212	93.582	Pass

Evaluating Flange on Nozzle [N3-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1588) + 4 \cdot \text{abs}(857) \cdot 239.776 \leq \pi \cdot 239.776^3 [(108.604 - 50.212) + 500 \cdot 108.604]$$

$$= 0.26232E+05 \leq 0.48805E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(108.604, 0.500 \cdot 108.604 + 108.604 - 26232.2 / (\pi \cdot 239.776^3))$$

$$= \min(108.604, 102.334)$$

$$= 102.334 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N4-Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2294) + 4 \cdot \text{abs}(1030) \cdot 293.751 \leq \pi \cdot 293.751^3 [(108.604 - 50.212) + 300 \cdot 108.604]$$

$$= 0.37907E+05 \leq 0.72444E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(108.604, 0.300 \cdot 108.604 + 108.604 - 37907.0 / (\pi \cdot 293.751^3))$$

$$= \min(108.604, 93.582)$$

$$= 93.582 \text{ kgf/cm}^2$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020
Basic Wind Speed for IS-875	180 km/hr
Wind Zone Number	5
Base Elevation	30 cm
Percent Wind for Hydrotest	33.0
Risk Factor [k1]	1.0
Terrain Category	2
Equipment Class	Class A
Topography Factor [k3]	1.0
Importance Factor [k4]	1.15
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Use Gust Response Factor (Dynamic Analysis)	No

Calculated Values:

Vessel Length to Diameter ratio: 5.590

Design Wind Speed [Vz]:
 = Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)
 = 50 • 1 • 1 • 1 • 1.15
 = 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:
 = max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)
 = max(0.7 • 1984, 1 • 0.93 • 1 • 0.6 • 57.5²)
 = 1849.9 N/m²

Element Wind Load (operating condition):
 = Wind Area • Cf • pd
 = 0.86 • 0.81 • 1850
 = 1286.9 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	1.45	1.0000	1.0000	1.0000	57.50	0.8118
20	1.45	1.0000	1.0000	1.0000	57.50	0.8118
30	1.45	1.0000	1.0000	1.0000	57.50	0.8118
40	1.45	1.0000	1.0000	1.0000	57.50	0.8118
60	1.45	1.0000	1.0000	1.0000	57.50	0.8118
70	1.45	1.0000	1.0000	1.0000	57.50	0.8118

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
10	20	145	267.788	8569.2	188.72	144.357
20	30	145	188.058	8180.52	188.72	137.809
30	40	145	222.48	40046.4	188.72	674.623
40	60	145	202.8	5962.32	188.72	100.441
60	70	145	208.56	120965	188.72	2037.78
70	80	145	208.08	9119.38	188.72	153.625

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite 28 Licensee: SL User
FileName : 401-E-31002-AK
Wind Load Calculation:

Step: 15 1:29pm

Page 60 of 162
Jul 4, 2026



HEXAGON

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1) \cdot 1$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 5224 \cdot 0.094 \cdot 1$
 $= 490.3 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.094 \cdot 41791 \cdot 1$
 $= 2614.9 \text{ kgf}$

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height cm	Weight kgf	Ope Load kgf
10	20	83	5223.83	490.288
20	30	55.3	5223.83	490.288
30	40	55	5223.83	490.288
40	60	55	5223.83	490.288
60	Sad1	55	5223.83	490.288
Sad1	70	55	5223.83	490.288
60	70	55	5223.83	490.288
70	80	55	5223.83	490.288

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	603.535 cm
Center of Gravity of the Liquid	590.846 cm
Center of Gravity of the Insulation	455.270 cm
Center of Gravity of the Nozzles	251.078 cm
Center of Gravity of the Tubesheet(s)	269.817 cm
Center of Gravity of the Tubes	545.818 cm
Center of Gravity of the Baffles	593.555 cm
Center of Gravity of Bare Shell New and Cold	217.832 cm
Center of Gravity of Bare Shell Corroded	212.283 cm
Vessel CG in the Operating Condition	346.778 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	315.556 cm
Vessel CG in the Test Condition	339.093 cm

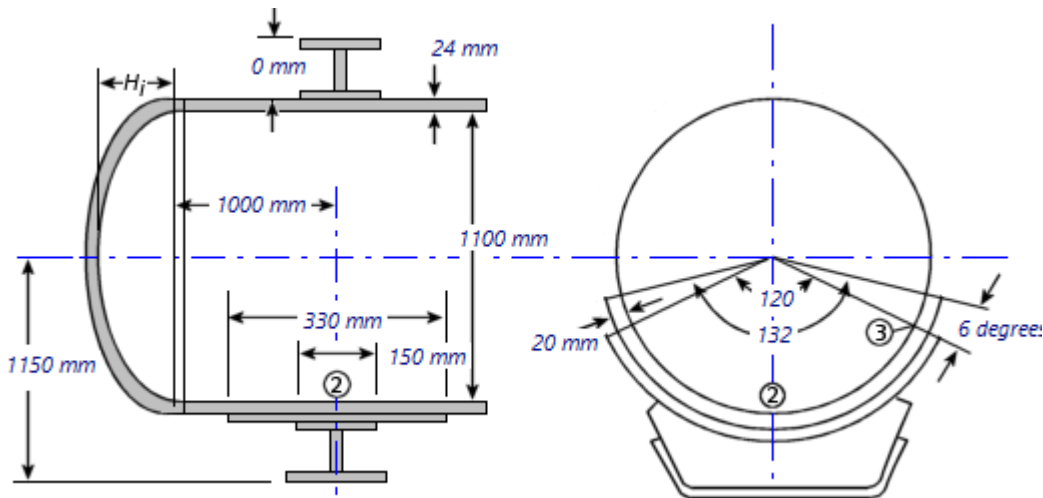
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	562.75	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	609.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1458.20	kgf/cm ²
Shell Allowable Stress used in Calculation		1458.19	kgf/cm ²
Head Allowable Stress used in Calculation		1325.61	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1150.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		57527.38	kgf
Pressure used in Analysis	P	50.022	kgf/cm ²

Horizontal Vessel Analysis Results:		
	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	484.35	1458.19
Long. Stress at Bottom of Midspan	766.77	1458.19
Long. Stress at Top of Saddles	958.48	1458.19
Long. Stress at Bottom of Saddles	441.03	1458.19
Tangential Shear in Shell	357.33	1166.55
Circ. Stress at Horn of Saddle	356.44	1822.74
Circ. Compressive Stress in Shell	31.61	1458.19

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr} (\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (3249/2 + 0) \cdot 1150/1029$$

$$= 5446.8 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(709, 0) \cdot 1150/3800$$

$$= 214.6 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(3922, 0) \cdot 1150/3800$$

$$= 1187.0 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr} (\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (3922/2 + 0) \cdot 1150/1029$$

$$= 6576.3 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 50951 + \max(215, 5447, 1187, 6576)$$

$$= 57527.4 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1851 (0 + \pi/4 (1.74 \cdot 1.2)^2)$$

$$= 6954.777 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	57905.81	kgf
Transverse Shear Load	1961.15	kgf
Longitudinal Shear Load	3922.30	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2 (1100/2 + 24 + 20) \sin(120/2), 2 \cdot 563)$$

$$= 1028.838 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$$

$$= -57527 \cdot 100 [1 - (1 - 100/609 + (56.3^2 - 0^2) / (2 \cdot 100 \cdot 609)) / (1 + (4 \cdot 0) / (3 \cdot 609))]$$

$$= -7945.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2 (\text{Rm}^2 - \text{hm}^2) / (L^2)) / (1 + (4\text{hm}^2) / (3L)) - 4a/L)$$

$$= 57527 \cdot 609 / 4 ((1 + 2 (56.3^2 - 0^2) / (609^2)) / (1 + (4 \cdot 0) / (3 \cdot 609)) - 4 \cdot 100/609)$$

$$= 31610.4 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm} / (2t) - M2 / (\pi \cdot \text{Rm}^2 t)$$

$$= 50 \cdot 563 / (2 \cdot 22.5) - 31610 / (\pi \cdot 563^2 \cdot 22.5)$$

$$= 484.35 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm} / (2t) + M2 / (\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 50 \cdot 563 / (2 \cdot 22.5) + 31610 / (\pi \cdot 563^2 \cdot 22.5)$$

$$= 766.77 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50 \cdot 563 / (2 \cdot 22.5) - 7945 / (0.11 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 958.48 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50 \cdot 563 / (2 \cdot 22.5) + 7945 / (0.19 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 441.03 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 57527 (609 - 2 \cdot 100) / (609 + (4 \cdot 0 / 3))$$

$$= 38647.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 38647 / (563 \cdot 22.5)$$

$$= 357.33 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 22.5)}$$

$$= 87.769 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 22.5)}, 2 \cdot 100)$$

$$= 325.54 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1458 / 1458, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 57527 \cdot 0.1 / (326(22.5 + 1 \cdot 20))$$

$$= -31.61 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -57527 / (4(22.5 + 1 \cdot 20)326) -$$

$$3 \cdot 0.053 \cdot 57527 / (2(22.5 + 1 \cdot 20)^2)$$

$$= -356.44 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3800.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.14190\text{E-}04 \cdot 3800 (450 - 21.1)$$

$$= 23.126 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2233.1	kgf/cm ²
Height of Web at Center	Hw,c	350.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(552 \cdot 22.5)} + 330)^3) / 12$
 $= 47.819 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 47.8 + 113 (240 - 11.3)^2$
 $= 59381.359 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	503.8	22.5	11.3	113.3	127518.2	0.594E+05
Wearplate	330.0	20.0	32.5	66.0	214500.0	0.285E+05
Web	20.0	524.0	304.5	104.8	3191160.2	0.283E+05
BasePlate	300.0	32.0	582.5	96.0	5592000.0	0.113E+06
Totals	...	...	...	380.1	9125178.0	0.229E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 9125178 / 380$
 $= 240.042 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 57527$
 $= 11708.0693 \text{ kgf}$

Tension Stress, St = (Fh/As) = 43.8834 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1339.8823 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1150 - 552 \cdot \sin(120/2) / 1.05$
 $= 693.913 \text{ mm}$

Bending Moment, M = Fh * d = 8124.3931 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 85.2230 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1488.7581 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(57527 + 378)300) / (4 \cdot 1200 \cdot 1489)}$
 $= 27.005 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (550 + 24 + 20)$
 $= 1028.838 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1029 / (4 - 1)$
 $= 342.946 \text{ mm}$



Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot Bpwid / 2 \\ &= 343 \cdot 300 / 2 \\ &= 514.419 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\ &= 57906 / (1200 \cdot 300) \\ &= 16.085 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 514 \cdot 16.1 \\ &= 8274.404 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= Rib \text{ Area} + Web \text{ Area} \\ &= 26 + 34.3 \\ &= 60.295 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 8274 / 60.3 \\ &= 137.233 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	150.0	...	30.0	...	562.

Rib dimension [D]:

$$\begin{aligned} &= Saddle \text{ Width} - Web \text{ Thickness} \\ &= 150 - 20 \\ &= 130.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0 / 60.3 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= Saddle \text{ Width} / 2 \\ &= 150 / 2 \\ &= 75.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\ &= \sqrt{ (562 / 60.3) } \\ &= 30.544 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk \\ &= 1150 - \cos(120 / 2) (550 + 24 + 20) - 32 \\ &= 821.000 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 2038900 / 2233) } \\ &= 134.247 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 821 / 30.5 \\ &= 26.879 \end{aligned}$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 3922 / (2 \cdot 1200) \cdot 343 \cdot 821 / 2$$

$$= 230.075 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (26.9 < 134) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (26.9)^2 / (2 \cdot 134^2)) 2233 / (5/3 + 3 \cdot (26.9) / (8 \cdot 134) - (26.9^3) / (8 \cdot 134^3))$$

$$= 1257 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 137 / 1257 + (230 \cdot 75 / 5624999) / 1489$$

$$= 0.315$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	130.0	0.0	26.0	0.0	561.
Web	342.9	20.0	0.0	68.6	0.0	22.9
Totals	...	...	...	94.6	...	584.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 94.6$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150 / 2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1150 - \sqrt{ (594 + 20)^2 - (343/2)^2 } - 32$$

$$= 549.288 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 584 / 94.6 }$$

$$= 24.848 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 549 / 24.8$$

$$= 22.106$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3922 / (2 \cdot 1200)$$

$$= 1.634 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.63 \cdot 343 \cdot 549$$

$$= 307.862 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 308 / 77871$$

$$= 395.351 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (22.1 < 134) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 134^2)) 2233 / (5/3 + 3 \cdot (22.1) / (8 \cdot 134) - (22.1^3) / (8 \cdot 134^3))$$

$$= 1275 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 138/1275 + (308 \cdot 75/584)/1489$$

$$= 0.374$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.02 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1458.20 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	51329.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1187.0 kgf
Maximum Transverse Force	Ft	1961.2 kgf
Maximum Longitudinal Force	F1	3922.3 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (Bolt Area \cdot Number of Bolts)$$

$$= 3922 / (4.23 \cdot 4)$$

$$= 231.5 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (Bolt Area \cdot Number of Bolts)$$

$$= 1961 / (4.23 \cdot 4)$$

$$= 115.8 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (BoltShearAllowable \cdot Nbolts)$
 = $3922 / (1100 \cdot 4)$
 = 0.8915 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1150 \cdot 1961 + 0$$

$$= 2255.33 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 2255/51329$$

$$= 43.94 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.89, 0]$$

$$= 0.8915 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 20, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.83$, Temp. Reduction = 9 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	562.75	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	609.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1458.20	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.67	cm
Shell Allowable Stress used in Calculation		1458.19	kgf/cm ²
Head Allowable Stress used in Calculation		1458.19	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1150.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		14979.90	kgf
Pressure used in Analysis	P	50.022	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		595.33	1458.19
Long. Stress at Bottom of Midspan		655.78	1458.19
Long. Stress at Top of Saddles		748.17	1458.19
Long. Stress at Bottom of Saddles		557.60	1458.19
Tangential Shear in Shell		87.55	1166.55
Circ. Stress at Horn of Saddle		92.82	1822.74
Circ. Compressive Stress in Shell		8.23	1458.19

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3249/2 + 0) \cdot 1150/1029$
 $= 5446.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(709, 0) \cdot 1150/3800$
 $= 214.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(3922, 0) \cdot 1150/3800$
 $= 1187.0 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3922/2 + 0) \cdot 1150/1029$
 $= 6576.3 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 8404 + \max(215, 5447, 1187, 6576)$$

$$= 14979.9 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1851 (0 + \pi/4(1.74 \cdot 1.2)^2)$$

$$= 6954.777 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	15358.34	kgf
Transverse Shear Load	1961.15	kgf
Longitudinal Shear Load	3922.30	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1100/2 + 24 + 20)\sin(120/2), 2 \cdot 563)$$

$$= 1028.838 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$$

$$= -14980 \cdot 100 [1 - (1 - 100/609 + (56.3^2 - 28.7^2)/(2 \cdot 100 \cdot 609)) / (1 + (4 \cdot 28.7)/(3 \cdot 609))]$$

$$= -2926.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4a/L)$$

$$= 14980 \cdot 609/4 ((1 + 2(56.3^2 - 28.7^2)/(609^2)) / (1 + (4 \cdot 28.7)/(3 \cdot 609)) - 4 \cdot 100/609)$$

$$= 6765.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 t)$$

$$= 50 \cdot 563/(2 \cdot 22.5) - 6766/(\pi \cdot 563^2 \cdot 22.5)$$

$$= 595.33 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 50 \cdot 563/(2 \cdot 22.5) + 6766/(\pi \cdot 563^2 \cdot 22.5)$$

$$= 655.78 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot \text{Rm}/(2t) - M1/(K1 \cdot \pi \cdot \text{Rm}^2 t)$$

$$= 50 \cdot 563/(2 \cdot 22.5) - 2926/(0.11 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 748.17 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot \text{Rm}/(2t) + M1/(K1^* \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 50 \cdot 563/(2 \cdot 22.5) + 2926/(0.19 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 557.60 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot \text{hm}/3))$$

$$= 14980 (609 - 2 \cdot 100)/(609 + (4 \cdot 28.7/3))$$

$$= 9469.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 9470 / (563 \cdot 22.5)$$

$$= 87.55 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 22.5)}$$

$$= 87.769 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 22.5)}, 2 \cdot 100)$$

$$= 325.54 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1458/1458, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 14980 \cdot 0.1 / (326(22.5 + 1 \cdot 20))$$

$$= -8.23 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -14980 / (4(22.5 + 1 \cdot 20)326) -$$

$$3 \cdot 0.053 \cdot 14980 / (2(22.5 + 1 \cdot 20)^2)$$

$$= -92.82 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2233.1	kgf/cm ²
Height of Web at Center	Hw,c	350.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 22.5)} + 330)22.5^3) / 12$$

$$= 47.819 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 47.8 + 113(240 - 11.3)^2$$

$$= 59381.359 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	503.8	22.5	11.3	113.3	127518.2	0.594E+05
Wearplate	330.0	20.0	32.5	66.0	214500.0	0.285E+05
Web	20.0	524.0	304.5	104.8	3191160.2	0.283E+05
BasePlate	300.0	32.0	582.5	96.0	5592000.0	0.113E+06
Totals	...	...	...	380.1	9125178.0	0.229E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 9125178/380$$

$$= 240.042 \text{ mm}$$



Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 14980 \\ &= 3048.7354 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 11.4271 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1339.8823 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1150 - 552 \cdot \sin(120/2) / 1.05 \\ &= 693.913 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2115.5603 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 22.1917 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1488.7581 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(14980 + 378)300 / (4 \cdot 1200 \cdot 1489)} \\ &= 13.908 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (550 + 24 + 20) \\ &= 1028.838 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1029 / (4 - 1) \\ &= 342.946 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 343 \cdot 300 / 2 \\ &= 514.419 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 15358 / (1200 \cdot 300) \\ &= 4.266 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 514 \cdot 4.27 \\ &= 2194.617 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 26 + 34.3 \\ &= 60.295 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 2195/60.3 \\ &= 36.398 \text{ kgf/cm}^2 \end{aligned}$$



Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	150.0	...	30.0	...	562.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 150 - 20
 = 130.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/60.3
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 150/2
 = 75.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{562/60.3}$
 = 30.544 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1150 - cos(120/2)(550 + 24 + 20) - 32
 = 821.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2038903/2233}$
 = 134.247

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 821/30.5
 = 26.879

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 3922/ (2 • 1200) • 343 • 821/2
 = 230.075 kgf-m.

Compressive Allowable, KL/r < Cc (26.9 < 134) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (26.9)^2 / (2 \cdot 134^2)) 2233 / (5/3 + 3 \cdot (26.9) / (8 \cdot 134) - (26.9)^3 / (8 \cdot 134^3))$
 = 1257 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 36.4/1257 + (230 • 75/5624999) / 1489
 = 0.235

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	130.0	0.0	26.0	0.0	561.
Web	342.9	20.0	0.0	68.6	0.0	22.9
Totals	...	...	...	94.6	...	584.

Distance to Centroid from Datum [ytot]:
 = AY / A



$$= 0/94.6$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1150 - \sqrt{(594 + 20)^2 - (343/2)^2} - 32$$

$$= 549.288 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(584/94.6)}$$

$$= 24.848 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 549/24.8$$

$$= 22.106$$

Unit Force [Force,u]:

$$= \text{FI} / (2 \cdot \text{Baseplate Length})$$

$$= 3922 / (2 \cdot 1200)$$

$$= 1.634 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.63 \cdot 343 \cdot 549$$

$$= 307.862 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 308/77871$$

$$= 395.351 \text{ kgf/cm}^2$$

Compressive Allowable, $\text{KL}/r < C_c$ ($22.1 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 134^2)) 2233 /$$

$$(5/3 + 3 \cdot (22.1) / (8 \cdot 134) - (22.1)^3 / (8 \cdot 134^3))$$

$$= 1275 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Mbase,c} \cdot \text{C1}/I) / \text{Sba}$$

$$= 36.7/1275 + (308 \cdot 75/584) / 1489$$

$$= 0.294$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.02 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1458.20 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	8782.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1187.0 kgf
Maximum Transverse Force	Ft	1961.2 kgf
Maximum Longitudinal Force	F1	3922.3 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 15398.8 \text{ kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $15399 / (4.23 \cdot 4)$
= 909.0 kgf/cm^2 . Must be less than 1100.0 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $1961 / (4.23 \cdot 4)$
= 115.8 kgf/cm^2 . Must be less than 1100.0 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $3922 / (1100 \cdot 4)$
= 0.8915 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1150 \cdot 1961 + 0$
= 2255.33 kgf-m .

Eccentricity (e):
= R_{mom} / QO
= $2255 / 8782$
= $256.81 \text{ mm} > B_{plen} / 6$ --> Uplift in Transverse direction

$f = B_{plen} / 2 - \text{Edgedis}$
= $1200 / 2 - 50.8$
= 549.20 mm

Modular Ratio Of Steel/Concrete (n1):
= E_S / E_C
= $2074056 / 219500$
= 9.45

$K1 = 3(e - 0.5 \cdot B_{plen})$
= $3(257 - 0.5 \cdot 1200)$
= -1029.56 mm

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
= $6 \cdot 9.45 \cdot 8.47 / 300 \cdot (549 + 257)$
= 129013.88 mm^2

$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$
= $-129014 \cdot (0.5 \cdot 1200 + 549)$
= $-148262745.83 \text{ mm}^3$

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1030 \cdot Y^2 + 129014 \cdot Y + -148262752 = 0$
 $Y = 1042.26 \text{ mm}$

$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$
= $(1200 / 2 - 1042 / 3 - 257)$
= -4.23

$\text{Denom} = (B_{plen} / 2 - Y / 3 + f)$
= $(1200 / 2 - 1042 / 3 + 549)$
= 801.78

Total Bolt Tension Force [Tforce]:
= $-QO \cdot \text{Num} / \text{Denom}$
= $-8782 \cdot -4.23 / 802$
= 46.37 kgf



Bolt Area Required due to Transverse Load [Bltareart]:
= Tforce / (Stba • Nbt)
= 46.4/(1400 • 2)
= 0.0166 cm²

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.89, 0.017]
= 0.8915 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= 2(QO + Tforce) / (Y • Bpwid)
= 2(8782 + 46.4)/(1042 • 300)
= 5.65 kgf/cm²

Distance from Baseplate Edge to the Web [ADIST]:
= (Bplen - WebIngh) / 2
= (1200 - 1149)/2
= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
= Tforce • Adist
= 46.4 • 2.54
= 1.18 kgf-m.

Equivalent Bearing Pressure (f1):
= fc • (Y - Adist) / Y
= 5.65 • (1042 - 25.4)/1042
= 5.51 kgf/cm²

Overturning Moment due to Bearing Pressure [Mc]:
= (Adist² • Bpwid / 6) • (f1 + 2 • fc)
= (25.4² • 300/6) • (5.51 + 2 • 5.65)
= 5.42 kgf-m.

Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(1.18, 5.42) / (300 \cdot 2187))}$
= 2.2263 mm

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 20, c = 1.5 mm, E* = 1
Thickness Ratio = tr • E*/(tg - c) = 0.83, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	3 °C

Review notes about nozzle loadings and supports in the Warnings report.

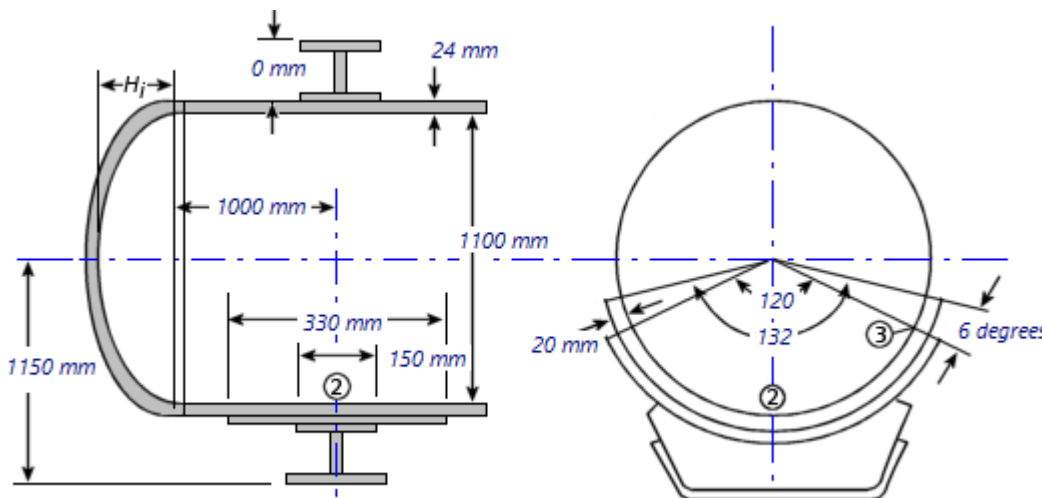
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	562.75	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	609.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1458.20	kgf/cm ²
Shell Allowable Stress used in Calculation		3002.91	kgf/cm ²
Head Allowable Stress used in Calculation		3002.91	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1150.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		10224.06	kgf
Pressure used in Analysis	P	67.611	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		820.41	3002.91
Long. Stress at Bottom of Midspan		870.61	3002.91
Long. Stress at Top of Saddles		904.68	3002.91
Long. Stress at Bottom of Saddles		812.71	3002.91
Tangential Shear in Shell		63.51	1684.79
Circ. Stress at Horn of Saddle		102.49	4504.36
Circ. Compressive Stress in Shell		7.41	3002.91

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$\begin{aligned} &= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\ &= 3 (1072/2 + 0) \cdot 1150/1029 \\ &= 1797.4 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Saddle Reaction Force [Fwl]:} \\ &= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls \\ &= \max(234, 0) \cdot 1150/3800 \\ &= 70.8 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Load Combination Results for Q + Wind or Seismic [Q]:} \\ &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\ &= 8427 + \max(70.8, 1797, 0, 0) \\ &= 10224.1 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Wind Force [FI]:} \\ &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\ &= 1.1 \cdot 1851 (0 + \pi/4(1.74 \cdot 1.2)^2) \\ &= 6954.777 \text{ N} \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	10602.50	kgf
Transverse Shear Load	536.02	kgf
Longitudinal Shear Load	234.03	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

$$\begin{aligned} &\text{Saddle Dimension [E]:} \\ &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm) \\ &= \min(2(1100/2 + 24 + 20)\sin(120/2), 2 \cdot 563) \\ &= 1028.838 \text{ mm} \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

$$\begin{aligned} &\text{Moment per Equation 4.15.1 [M1]:} \\ &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\ &= -10224 \cdot 100 [1 - (1 - 100/609 + (56.3^2 - 0^2)/(2 \cdot 100 \cdot 609))/(1 + (4 \cdot 0)/(3 \cdot 609))] \\ &= -1412.1 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Moment per Equation 4.15.2 [M2]:} \\ &= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\ &= 10224 \cdot 609/4((1 + 2(56.3^2 - 0^2)/(609^2))/(1 + (4 \cdot 0)/(3 \cdot 609)) - 4 \cdot 100/609) \\ &= 5618.0 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Top of Shell (4.15.4) [\sigma1]:} \\ &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\ &= 67.6 \cdot 563/(2 \cdot 22.5) - 5618/(\pi \cdot 563^2 \cdot 22.5) \\ &= 820.41 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Bottom of Shell (4.15.5) [\sigma2]:} \\ &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\ &= 67.6 \cdot 563/(2 \cdot 22.5) + 5618/(\pi \cdot 563^2 \cdot 22.5) \\ &= 870.61 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Top of Shell at Support (4.15.8) [\sigma*3]:} \\ &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\ &= 67.6 \cdot 563/(2 \cdot 22.5) - 1412/(0.11 \cdot \pi \cdot 563^2 \cdot 22.5) \\ &= 904.68 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) [\sigma*4]:} \\ &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \end{aligned}$$

$$= 67.6 \cdot 563 / (2 \cdot 22.5) + 1412 / (0.19 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 812.71 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 10224 (609 - 2 \cdot 100) / (609 + (4 \cdot 0/3))$$

$$= 6868.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 6869 / (563 \cdot 22.5)$$

$$= 63.51 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 22.5)}$$

$$= 87.769 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 22.5)}, 2 \cdot 100)$$

$$= 325.54 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1458/3003, 1)$$

$$= 0.4856$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 10224 \cdot 0.1 / (326(22.5 + 0.49 \cdot 20))$$

$$= -7.41 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q/(2(t + \eta \cdot tr)^2)$$

$$= -10224/(4(22.5 + 0.49 \cdot 20)326) -$$

$$3 \cdot 0.053 \cdot 10224/(2(22.5 + 0.49 \cdot 20)^2)$$

$$= -102.49 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	3161.0	kgf/cm ²
Height of Web at Center	Hw,c	350.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 22.5)} + 330)22.5^3)/12$$

$$= 47.819 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 47.8 + 113(240 - 11.3)^2$$

$$= 59381.359 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	503.8	22.5	11.3	113.3	127518.2	0.594E+05
Wearplate	330.0	20.0	32.5	66.0	214500.0	0.285E+05
Web	20.0	524.0	304.5	104.8	3191160.2	0.283E+05
BasePlate	300.0	32.0	582.5	96.0	5592000.0	0.113E+06



Totals	...	...	...	380.1	9125178.0	0.229E+06
--------	-----	-----	-----	-------	-----------	-----------

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 9125178/380 \\ &= 240.042 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 10224 \\ &= 2080.8188 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 7.7992 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1896.5723 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1150 - 552 \cdot \sin(120/2) / 1.05 \\ &= 693.913 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1443.9093 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 15.1463 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 2107.3025 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\ &= \sqrt{(3(10224 + 378) \cdot 300) / (4 \cdot 1200 \cdot 2107)} \\ &= 9.713 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (550 + 24 + 20) \\ &= 1028.838 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1029 / (4 - 1) \\ &= 342.946 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 343 \cdot 300/2 \\ &= 514.419 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 10602 / (1200 \cdot 300) \\ &= 2.945 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 514 \cdot 2.95 \\ &= 1515.035 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 26 + 34.3 \end{aligned}$$



$$= 60.295 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1515/60.3$$

$$= 25.127 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	150.0	...	30.0	...	562.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150 - 20$$

$$= 130.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(562/60.3)}$$

$$= 30.544 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1150 - \cos(120/2) (550 + 24 + 20) - 32$$

$$= 821.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/3161)}$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 821/30.5$$

$$= 26.879$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 234 / (2 \cdot 1200) \cdot 343 \cdot 821/2$$

$$= 13.728 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($26.9 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (26.9)^2 / (2 \cdot 113^2)) 3161 /$$

$$(5/3 + 3 \cdot (26.9) / (8 \cdot 113) - (26.9^3) / (8 \cdot 113^3))$$

$$= 1751 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 25.1/1751 + (13.7 \cdot 75/5624999) / 2107$$

$$= 0.023$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	20.0	130.0	0.0	26.0	0.0	561.
Web	342.9	20.0	0.0	68.6	0.0	22.9
Totals	...	...	...	94.6	...	584.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/94.6$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1150 - \sqrt{(594 + 20)^2 - (343/2)^2} - 32$$

$$= 549.288 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(584/94.6)}$$

$$= 24.848 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 549/24.8$$

$$= 22.106$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 234 / (2 \cdot 1200)$$

$$= 0.098 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.098 \cdot 343 \cdot 549$$

$$= 18.369 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 18.4/77871$$

$$= 23.589 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($22.1 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 113^2)) 3161 /$$

$$(5/3 + 3 \cdot (22.1) / (8 \cdot 113) - (22.1)^3 / (8 \cdot 113^3))$$

$$= 1783 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 25.4/1783 + (18.4 \cdot 75/584) / 2107$$

$$= 0.025$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1509.10 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	8805.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	70.8 kgf
Maximum Transverse Force	Ft	536.0 kgf

Maximum Longitudinal Force F1 234.0 kgf
 Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 234 / (4.23 \cdot 4)$
 $= 13.8 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 536 / (4.23 \cdot 4)$
 $= 31.6 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 234 / (1052 \cdot 4)$
 $= 0.0556 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1150 \cdot 536 + 0$
 $= 616.43 \text{ kgf-m}$.

Eccentricity (e):
 $= R_{mom} / QO$
 $= 616/8805$
 $= 70.01 \text{ mm} < B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 0.056, 0]$
 $= 0.0556 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	562.75	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	609.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1458.20	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.67	cm
Shell Allowable Stress used in Calculation		3002.91	kgf/cm ²
Head Allowable Stress used in Calculation		3002.91	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1150.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 10219.02 kgf
 Pressure used in Analysis P 67.611 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	824.89	3002.91
Long. Stress at Bottom of Midspan	866.13	3002.91
Long. Stress at Top of Saddles	929.15	3002.91
Long. Stress at Bottom of Saddles	799.15	3002.91
Tangential Shear in Shell	59.73	1684.79
Circ. Stress at Horn of Saddle	102.44	4504.36
Circ. Compressive Stress in Shell	7.41	3002.91

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3 (1072/2 + 0) \cdot 1150/1029$
 $= 1797.4 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(234, 0) \cdot 1150/3800$
 $= 70.8 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 8422 + \max(70.8, 1797, 0, 0)$
 $= 10219.0 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1851 (0 + \pi/4(1.74 \cdot 1.2)^2)$
 $= 6954.777 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 10597.45 kgf
 Transverse Shear Load 536.02 kgf
 Longitudinal Shear Load 234.03 kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1100/2 + 24 + 20) \sin(120/2), 2 \cdot 563)$
 $= 1028.838 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L)) / (1 + (4\text{hm})/3L)]$
 $= -10219 \cdot 100 [1 - (1 - 100/609 + (56.3^2 - 28.7^2) / (2 \cdot 100 \cdot 609)) / (1 + (4 \cdot 28.7)/(3 \cdot 609))]$
 $= -1996.2 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 10219 \cdot 609 / 4 \cdot ((1 + 2(56.3^2 - 28.7^2) / (609^2)) / (1 + (4 \cdot 28.7) / (3 \cdot 609)) - 4 \cdot 100 / 609) \\
 &= 4615.5 \text{ kgf-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 67.6 \cdot 563 / (2 \cdot 22.5) - 4616 / (\pi \cdot 563^2 \cdot 22.5) \\
 &= 824.89 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 67.6 \cdot 563 / (2 \cdot 22.5) + 4616 / (\pi \cdot 563^2 \cdot 22.5) \\
 &= 866.13 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 67.6 \cdot 563 / (2 \cdot 22.5) - 1996 / (0.11 \cdot \pi \cdot 563^2 \cdot 22.5) \\
 &= 929.15 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 67.6 \cdot 563 / (2 \cdot 22.5) + 1996 / (0.19 \cdot \pi \cdot 563^2 \cdot 22.5) \\
 &= 799.15 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 10219 \cdot (609 - 2 \cdot 100) / (609 + (4 \cdot 28.7 / 3)) \\
 &= 6459.9 \text{ kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (\pi \cdot Rm \cdot t) \\
 &= 1.17 \cdot 6460 / (\pi \cdot 563 \cdot 22.5) \\
 &= 59.73 \text{ kgf/cm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{Rm \cdot t} \\
 &= 0.78 \cdot \sqrt{563 \cdot 22.5} \\
 &= 87.769 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\
 &= \min(150 + 1.56 \cdot \sqrt{563 \cdot 22.5}, 2 \cdot 100) \\
 &= 325.54 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(1458 / 3003, 1) \\
 &= 0.4856
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 10219 \cdot 0.1 / (326(22.5 + 0.49 \cdot 20)) \\
 &= -7.41 \text{ kgf/cm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -10219 / (4(22.5 + 0.49 \cdot 20)326) - 3 \cdot 0.053 \cdot 10219 / (2(22.5 + 0.49 \cdot 20)^2) \\
 &= -102.44 \text{ kgf/cm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	

Saddle Yield Stress Sy 3161.0 kgf/cm²
 Height of Web at Center Hw,c 350.0 mm
 Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{(552 \cdot 22.5)} + 330) 22.5^3) / 12$
 $= 47.819 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 47.8 + 113 (240 - 11.3)^2$
 $= 59381.359 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	503.8	22.5	11.3	113.3	127518.2	0.594E+05
Wearplate	330.0	20.0	32.5	66.0	214500.0	0.285E+05
Web	20.0	524.0	304.5	104.8	3191160.2	0.283E+05
BasePlate	300.0	32.0	582.5	96.0	5592000.0	0.113E+06
Totals	...	...	...	380.1	9125178.0	0.229E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 9125178 / 380$
 $= 240.042 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 10219$
 $= 2079.7917 \text{ kgf}$

Tension Stress, St = (Fh/As) = 7.7953 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1896.5723 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1150 - 552 \cdot \sin(120/2) / 1.05$
 $= 693.913 \text{ mm}$

Bending Moment, M = Fh * d = 1443.1965 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 15.1388 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 2107.3025 kgf/cm²

Minimum Thickness of Baseplate per Moss:
 $= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(10219 + 378) 300) / (4 \cdot 1200 \cdot 2107)}$
 $= 9.710 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (550 + 24 + 20)$
 $= 1028.838 \text{ mm}$

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1029/(4 - 1)
 = 342.946 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 343 • 300/2
 = 514.419 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 10597/(1200 • 300)
 = 2.944 kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 514 • 2.94
 = 1514.314 kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 26 + 34.3
 = 60.295 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 1514/60.3
 = 25.115 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	150.0	...	30.0	...	562.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 150 - 20
 = 130.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/60.3
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 150/2
 = 75.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{562/60.3}$
 = 30.544 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1150 - cos(120/2)(550 + 24 + 20) - 32
 = 821.000 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038903/3161}$
 = 112.838

Slenderness ratio [KL/r]:
 = KL/r

$$= 1 \cdot 821/30.5$$

$$= 26.879$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 234 / (2 \cdot 1200) \cdot 343 \cdot 821/2$$

$$= 13.728 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($26.9 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (26.9)^2 / (2 \cdot 113^2)) 3161 /$$

$$(5/3 + 3 \cdot (26.9) / (8 \cdot 113) - (26.9^3) / (8 \cdot 113^3))$$

$$= 1751 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 25.1/1751 + (13.7 \cdot 75/5624999) / 2107$$

$$= 0.023$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	130.0	0.0	26.0	0.0	561.
Web	342.9	20.0	0.0	68.6	0.0	22.9
Totals	...	...	...	94.6	...	584.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/94.6$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1150 - \sqrt{ (594 + 20)^2 - (343/2)^2 } - 32$$

$$= 549.288 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (584/94.6) }$$

$$= 24.848 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 549/24.8$$

$$= 22.106$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 234 / (2 \cdot 1200)$$

$$= 0.098 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.098 \cdot 343 \cdot 549$$

$$= 18.369 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 18.4/77871$$

$$= 23.589 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($22.1 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 113^2)) \cdot 3161 / (5/3 + 3 \cdot (22.1) / (8 \cdot 113) - (22.1^3) / (8 \cdot 113^3))$$

$$= 1783 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 25.3/1783 + (18.4 \cdot 75/584) / 2107$$

$$= 0.025$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1509.10 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	8800.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	70.8 kgf
Maximum Transverse Force	Ft	536.0 kgf
Maximum Longitudinal Force	F1	234.0 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 234 / (4.23 \cdot 4)$$

$$= 13.8 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 536 / (4.23 \cdot 4)$$

$$= 31.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 234 / (1052 \cdot 4)$$

$$= 0.0556 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1150 \cdot 536 + 0$$

$$= 616.43 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 616/8800$$

$$= 70.05 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.056, 0]$$

$$= 0.0556 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

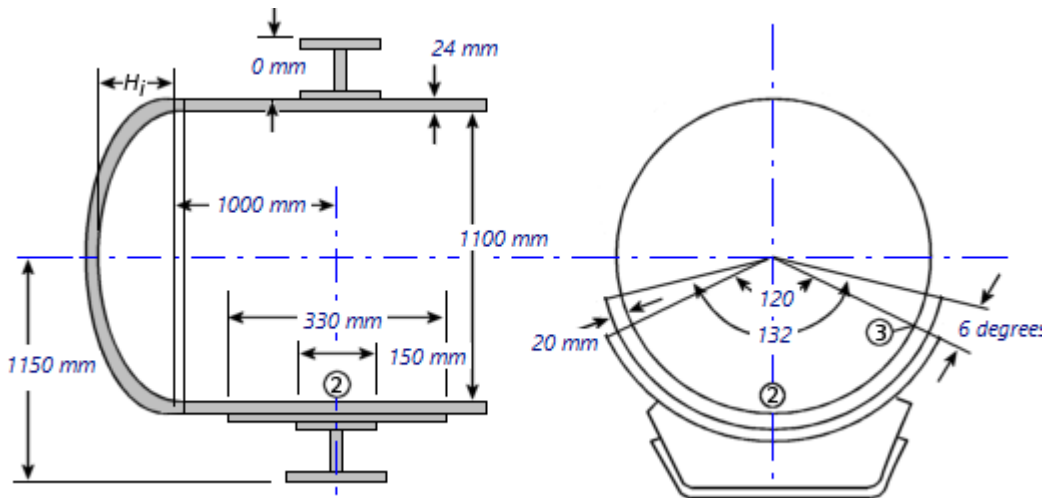
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	562.75	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	609.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1458.20	kgf/cm ²
Shell Allowable Stress used in Calculation		1458.19	kgf/cm ²
Head Allowable Stress used in Calculation		1325.61	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1150.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	57527.38	kgf
Pressure used in Analysis	P	-1.055 kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-154.40	-154.40	-2009.82
Long. Stress at Top of Midspan	154.40	154.40	1458.19
Long. Stress at Bottom of Midspan	128.02	128.02	1458.19
Long. Stress at Top of Saddles	319.73	319.73	1458.19
Long. Stress at Bottom of Saddles	-197.71	-197.71	-2009.82
Long. Stress at Bottom of Saddles	197.71	197.71	1458.19
Tangential Shear in Shell	357.33	357.33	1166.55
Circ. Stress at Horn of Saddle	356.44	356.44	1822.74
Circ. Compressive Stress in Shell	31.61	31.61	1458.19

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (3249/2 + 0) \cdot 1150/1029$
 $= 5446.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(709, 0) \cdot 1150/3800$
 $= 214.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(3922, 0) \cdot 1150/3800$
 $= 1187.0 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (3922/2 + 0) \cdot 1150/1029$
 $= 6576.3 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 50951 + \max(215, 5447, 1187, 6576)$
 $= 57527.4 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1851 (0 + \pi/4 (1.74 \cdot 1.2)^2)$
 $= 6954.777 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	57905.81	kgf
Transverse Shear Load	1961.15	kgf
Longitudinal Shear Load	3922.30	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2 (1100/2 + 24 + 20) \sin(120/2), 2 \cdot 563)$
 $= 1028.838 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$
 $= -57527 \cdot 100 [1 - (1 - 100/609 + (56.3^2 - 0^2) / (2 \cdot 100 \cdot 609)) / (1 + (4 \cdot 0) / (3 \cdot 609))]$
 $= -7945.2 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2 (\text{Rm}^2 - \text{hm}^2) / (L^2)) / (1 + (4\text{hm}^2) / (3L)) - 4a/L)$
 $= 57527 \cdot 609 / 4 ((1 + 2 (56.3^2 - 0^2) / (609^2)) / (1 + (4 \cdot 0) / (3 \cdot 609)) - 4 \cdot 100 / 609)$
 $= 31610.4 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm} / (2t) - M2 / (\pi \cdot \text{Rm}^2 t)$
 $= -1.05 \cdot 563 / (2 \cdot 22.5) - 31610 / (\pi \cdot 563^2 \cdot 22.5)$
 $= -154.40 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 563 / (2 \cdot 22.5) + 31610 / (\pi \cdot 563^2 \cdot 22.5)$$

$$= 128.02 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 563 / (2 \cdot 22.5) - 7945 / (0.11 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 319.73 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 563 / (2 \cdot 22.5) + 7945 / (0.19 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= -197.71 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 57527 (609 - 2 \cdot 100) / (609 + (4 \cdot 0 / 3))$$

$$= 38647.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 38647 / (563 \cdot 22.5)$$

$$= 357.33 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 22.5)}$$

$$= 87.769 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 22.5)}, 2 \cdot 100)$$

$$= 325.54 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1458 / 1458, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 57527 \cdot 0.1 / (326(22.5 + 1 \cdot 20))$$

$$= -31.61 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [$\sigma_{7,r}$]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -57527 / (4(22.5 + 1 \cdot 20)326) -$$

$$3 \cdot 0.053 \cdot 57527 / (2(22.5 + 1 \cdot 20)^2)$$

$$= -356.44 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2233.1	kgf/cm ²
Height of Web at Center	Hw,c	350.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 22.5)} + 330)^3) / 12$$

$$= 47.819 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 47.8 + 113 (240 - 11.3)²
 = 59381.359 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	503.8	22.5	11.3	113.3	127518.2	0.594E+05
Wearplate	330.0	20.0	32.5	66.0	214500.0	0.285E+05
Web	20.0	524.0	304.5	104.8	3191160.2	0.283E+05
BasePlate	300.0	32.0	582.5	96.0	5592000.0	0.113E+06
Totals	...	...	...	380.1	9125178.0	0.229E+06

Distance to Centroid [C1]:
 = AY / A
 = 9125178/380
 = 240.042 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.2 • 57527
 = 11708.0693 kgf

Tension Stress, St = (Fh/As) = 43.8834 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1339.8823 kgf/cm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1150 - 552 • sin(120/2)/1.05
 = 693.913 mm

Bending Moment, M = Fh * d = 8124.3931 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 85.2230 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1488.7581 kgf/cm²

Minimum Thickness of Baseplate per Moss:
 = $\sqrt{3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress)}$
 = $\sqrt{3(57527 + 378)300/(4 • 1200 • 1489)}$
 = 27.005 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120/2)(550 + 24 + 20)
 = 1028.838 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1029/(4 - 1)
 = 342.946 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 343 • 300/2
 = 514.419 cm²

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 57906 / (1200 \cdot 300)$
 $= 16.085 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 514 \cdot 16.1$
 $= 8274.404 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 26 + 34.3$
 $= 60.295 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 8274 / 60.3$
 $= 137.233 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	150.0	...	30.0	...	562.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 150 - 20$
 $= 130.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 60.3$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 150 / 2$
 $= 75.000 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{ (562 / 60.3)}$
 $= 30.544 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1150 - \cos(120 / 2) (550 + 24 + 20) - 32$
 $= 821.000 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{ (2 \cdot \pi^2 \cdot 2038900 / 2233)}$
 $= 134.247$

Slenderness ratio [KL/r]:
 $= KL / r$
 $= 1 \cdot 821 / 30.5$
 $= 26.879$

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 3922 / (2 \cdot 1200) \cdot 343 \cdot 821 / 2$
 $= 230.075 \text{ kgf-m.}$

Compressive Allowable, $KL/r < C_c$ ($26.9 < 134$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$



$$= (1 - (26.9)^2 / (2 \cdot 134^2)) 2233 / (5/3 + 3 \cdot (26.9) / (8 \cdot 134) - (26.9^3) / (8 \cdot 134^3))$$

$$= 1257 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 137 / 1257 + (230 \cdot 75 / 5624999) / 1489$$

$$= 0.315$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	130.0	0.0	26.0	0.0	561.
Web	342.9	20.0	0.0	68.6	0.0	22.9
Totals	...	...	...	94.6	...	584.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 94.6$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150 / 2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1150 - \sqrt{(594 + 20)^2 - (343/2)^2} - 32$$

$$= 549.288 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(584 / 94.6)}$$

$$= 24.848 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 549 / 24.8$$

$$= 22.106$$

Unit Force [Force,u]:

$$= F_i / (2 \cdot \text{Baseplate Length})$$

$$= 3922 / (2 \cdot 1200)$$

$$= 1.634 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.63 \cdot 343 \cdot 549$$

$$= 307.862 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 308 / 77871$$

$$= 395.351 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (22.1 < 134) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 134^2)) 2233 / (5/3 + 3 \cdot (22.1) / (8 \cdot 134) - (22.1^3) / (8 \cdot 134^3))$$

$$= 1275 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 138 / 1275 + (308 \cdot 75 / 584) / 1489$$

$$= 0.374$$

Input and Calculated Values:

Vessel Mean Radius	Rm	562.75	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	609.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	330.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	1458.20	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.67	cm
Shell Allowable Stress used in Calculation		1458.19	kgf/cm ²
Head Allowable Stress used in Calculation		1458.19	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1150.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		14979.90	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-43.41	-2009.82
Long. Stress at Top of Midspan	43.41	1458.19
Long. Stress at Bottom of Midspan	17.04	1458.19
Long. Stress at Top of Saddles	109.43	1458.19
Long. Stress at Bottom of Saddles	-81.15	-2009.82
Long. Stress at Bottom of Saddles	81.15	1458.19
Tangential Shear in Shell	87.55	1166.55
Circ. Stress at Horn of Saddle	92.82	1822.74
Circ. Compressive Stress in Shell	8.23	1458.19

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (3249/2 + 0) \cdot 1150/1029$
 $= 5446.8\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(709, 0) \cdot 1150/3800$
 $= 214.6\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(3922, 0) \cdot 1150/3800$
 $= 1187.0\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (3922/2 + 0) \cdot 1150/1029$
 $= 6576.3\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$



$$= 8404 + \max(215, 5447, 1187, 6576)$$

$$= 14979.9 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1851(0 + \pi/4(1.74 \cdot 1.2)^2)$$

$$= 6954.777 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	15358.34	kgf
Transverse Shear Load	1961.15	kgf
Longitudinal Shear Load	3922.30	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1100/2 + 24 + 20)\sin(120/2), 2 \cdot 563)$$

$$= 1028.838 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -14980 \cdot 100 [1 - (1 - 100/609 + (56.3^2 - 28.7^2)/(2 \cdot 100 \cdot 609))/(1 + (4 \cdot 28.7)/(3 \cdot 609))]$$

$$= -2926.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 14980 \cdot 609/4((1 + 2(56.3^2 - 28.7^2)/(609^2))/(1 + (4 \cdot 28.7)/(3 \cdot 609)) - 4 \cdot 100/609)$$

$$= 6765.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= -1.05 \cdot 563/(2 \cdot 22.5) - 6766/(\pi \cdot 563^2 \cdot 22.5)$$

$$= -43.41 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 563/(2 \cdot 22.5) + 6766/(\pi \cdot 563^2 \cdot 22.5)$$

$$= 17.04 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= -1.05 \cdot 563/(2 \cdot 22.5) - 2926/(0.11 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= 109.43 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.05 \cdot 563/(2 \cdot 22.5) + 2926/(0.19 \cdot \pi \cdot 563^2 \cdot 22.5)$$

$$= -81.15 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 14980(609 - 2 \cdot 100)/(609 + (4 \cdot 28.7/3))$$

$$= 9469.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 9470 / (563 \cdot 22.5)$$

$$= 87.55 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 22.5)}$$

$$= 87.769 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 22.5)}, 2 \cdot 100)$$

$$= 325.54 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1458/1458, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 14980 \cdot 0.1 / (326 (22.5 + 1 \cdot 20))$$

$$= -8.23 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -14980 / (4(22.5 + 1 \cdot 20)326) -$$

$$3 \cdot 0.053 \cdot 14980 / (2(22.5 + 1 \cdot 20)^2)$$

$$= -92.82 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2233.1	kgf/cm ²
Height of Web at Center	Hw,c	350.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 22.5)} + 330)22.5^3) / 12$$

$$= 47.819 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 47.8 + 113 (240 - 11.3)^2$$

$$= 59381.359 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	503.8	22.5	11.3	113.3	127518.2	0.594E+05
Wearplate	330.0	20.0	32.5	66.0	214500.0	0.285E+05
Web	20.0	524.0	304.5	104.8	3191160.2	0.283E+05
BasePlate	300.0	32.0	582.5	96.0	5592000.0	0.113E+06
Totals	...	...	...	380.1	9125178.0	0.229E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 9125178/380$$

$$= 240.042 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$
$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$
$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.2 \cdot 14980$$
$$= 3048.7354 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.4271 \text{ kgf/cm}^2$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1339.8823 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$
$$= 1150 - 552 \cdot \sin(120/2) / 1.05$$
$$= 693.913 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2115.5603 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 22.1917 \text{ kgf/cm}^2$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1488.7581 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt{3(14980 + 378)300 / (4 \cdot 1200 \cdot 1489)}$$
$$= 13.908 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 120/2) (550 + 24 + 20)$$
$$= 1028.838 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$
$$= 1029 / (4 - 1)$$
$$= 342.946 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 343 \cdot 300 / 2$$
$$= 514.419 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 15358 / (1200 \cdot 300)$$
$$= 4.266 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 514 \cdot 4.27$$
$$= 2194.617 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 26 + 34.3$$
$$= 60.295 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$
$$= 2195 / 60.3$$
$$= 36.398 \text{ kgf/cm}^2$$

Check of Outside Ribs:



Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
Rib+Web 20.0	150.0	...	30.0	...	562.

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 150 - 20
 = 130.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/60.3
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 150/2
 = 75.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{562/60.3}$
 = 30.544 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1150 - cos(120/2)(550 + 24 + 20) - 32
 = 821.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038903/2233}$
 = 134.247

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 821/30.5
 = 26.879

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 3922 / (2 • 1200) • 343 • 821/2
 = 230.075 kgf-m.

Compressive Allowable, KL/r < Cc (26.9 < 134) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (26.9)² / (2 • 134²)) 2233 /
 (5/3 + 3 • (26.9) / (8 • 134) - (26.9³) / (8 • 134³))
 = 1257 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 36.4/1257 + (230 • 75/5624999) / 1489
 = 0.235

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
Rib 20.0	130.0	0.0	26.0	0.0	561.
Web 342.9	20.0	0.0	68.6	0.0	22.9
Totals ...	...	...	94.6	...	584.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/94.6
 = 0.000 mm



Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 150/2 \\ &= 75.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk} \\ &= 1150 - \sqrt{(594 + 20)^2 - (343/2)^2} - 32 \\ &= 549.288 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(584/94.6)} \\ &= 24.848 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 549/24.8 \\ &= 22.106 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} &= FI / (2 \cdot \text{Baseplate Length}) \\ &= 3922 / (2 \cdot 1200) \\ &= 1.634 \text{ kgf/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} &= \text{Unit Force} \cdot e \cdot L \\ &= 1.63 \cdot 343 \cdot 549 \\ &= 307.862 \text{ kgf-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} &= \text{Bending Moment} / \text{Section Modulus} \\ &= 308/77871 \\ &= 395.351 \text{ kgf/cm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($22.1 < 134$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (22.1)^2 / (2 \cdot 134^2)) 2233 / \\ &\quad (5/3 + 3 \cdot (22.1) / (8 \cdot 134) - (22.1^3) / (8 \cdot 134^3)) \\ &= 1275 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba} \\ &= 36.7/1275 + (308 \cdot 75/584)/1489 \\ &= 0.294 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: N1-Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	450	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	450	°C

Parent Material		SA-336 F11	
Parent Allowable Stress at Temperature	Sv	1325.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	2235.0000	mm
Shell Finished (Minimum) Thickness	t	82.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		165.82	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1325.61	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		450.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
-----------------	--	--------	--

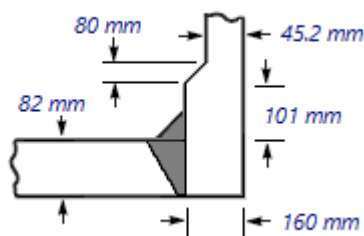
Hub Height of Integral Nozzle	h	101.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		2500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N1-Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 366.725 mm.
Actual Thickness Used in Calculation 45.237 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $163 \cdot 550 / (1326 \cdot 1 - 0.6 \cdot 163)$
= 73.0930 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $163 \cdot 185 / (1326 \cdot 1 - 0.6 \cdot 163)$
= 24.5676 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.1627 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 115/0.5773
= 198.7748 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

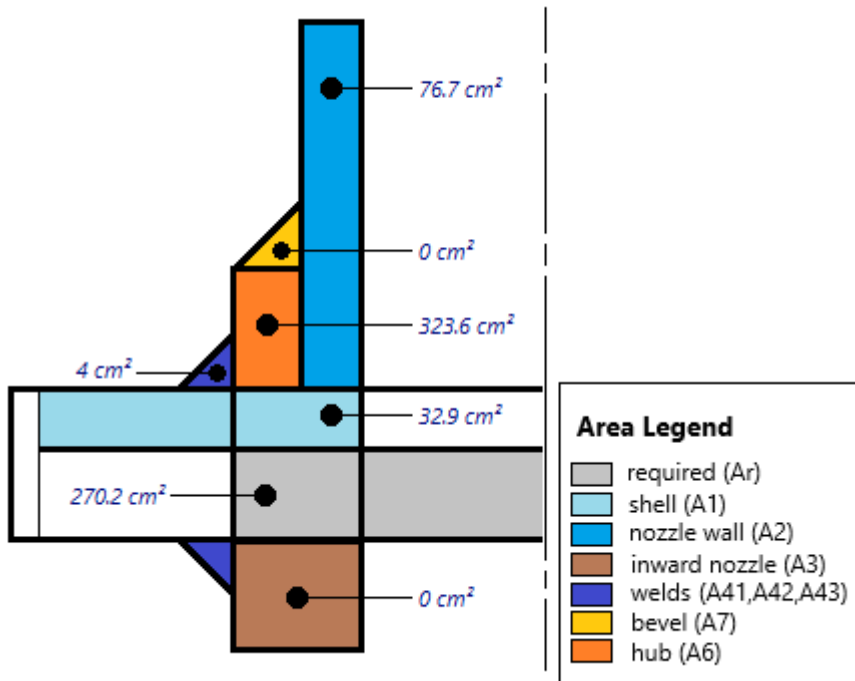
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.4504	mm
Parallel to Vessel Wall, opening length	d	369.7252	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	205.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1326/1326)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1326/1326)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	270.243	10.599	NA
Area in Shell	A1	32.931	281.977	NA
Area in Nozzle Wall	A2	76.679	170.299	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.000	4.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	323.631	323.631	NA
TOTAL AREA AVAILABLE	Atot	437.241	779.906	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (370 \cdot 73.1 \cdot 1 + 2 \cdot 158 \cdot 73.1 \cdot 1 \cdot (1 - 1))$
 $= 270.243 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 370(1 \cdot 82 - 1 \cdot 73.1) - 2 \cdot 158$
 $(1 \cdot 82 - 1 \cdot 73.1) \cdot (1 - 1)$
 $= 32.931 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho)(tn - trn)fr2$
 $= (2 \cdot 200)(43.7 - 24.6)1$
 $= 76.679 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 20^2 \cdot 1 + (0)^2 \cdot 1$
 $= 4.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(200, 200, 101)) \cdot (160 - 45.2) \cdot 1 \\ &= 323.631 \text{ cm}^2 \text{ Includes Bevel Area of } 91.810 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 26.0676 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 4.0000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 73.0930 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 73.0930 \text{ mm} \\ \text{Wall Thickness, shell/head, external pressure} & \quad t_{rb2} = 0.4378 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.0000 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 9.8312 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[9.83, \max(73.1, 4)] \\ &= 9.8312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(26.1, 9.83) \\ &= 26.0676 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 45.237 = 39.583 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 435.8,	Allowable	: 1325.6 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2980.2 kgf/cm ²	Passed
Occasional	: 308.3,	Allowable	: 1763.1 kgf/cm ²	Passed
Shear	: 86.4,	Allowable	: 927.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 39.6$, $t_r = 23$, $c = 1.5 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.6$, Temp. Reduction = 22°C

Min Metal Temp. w/o impact per UCS-66, Curve A	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	9 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 82$, $t_r = 68.5$, $c = 0 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9°C

Min Metal Temp. w/o impact per UCS-66, Curve B	27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	18 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $163.16/433.49 = 0.376$

Weld Size Calculations, Description: N1-Tube inlet

Intermediate Calc. for nozzle/shell welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (270 - 32.9 + 2 • 158 • 1 •
(1 • 82 - 73.1))1326)
= 352010.47 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (400 + 4 - 0 • 1) • 1326
= 535955.19 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (400 + 0 + 4 + (260)) • 1326
= 880533.00 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (400 + 0 + 4 + 0 + (260)) • 1326
= 880533.00 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 687 • 20 • 0.49 • 1326
= 140134. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 264) • (160 - 1.5) • 0.7 • 1326
= 1220339. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 687 • (24 - 0) • 0.74 • 1326
= 253957. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (140134 + 1220338) = 1360472 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (140134 + 0 + 253957 + 0) = 394091 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (140134 + 253957 + 0) = 394091 kgf

Summary of Failure Path Calculations:

Path 1-1 = 1360472 kgf, must exceed W = 352010 kgf	or W1 = 535955 kgf
Path 2-2 = 394090 kgf, must exceed W = 352010 kgf	or W2 = 880533 kgf
Path 3-3 = 394090 kgf, must exceed W = 352010 kgf	or W3 = 880533 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 179 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 120.3465 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 402.3465 mm

Input Echo, WRC297 Item 1, Description: N1-Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	82.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-336 F11	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1325.61	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	366.725	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1325.61	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-4818.42	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	3671.16	kgf
Circumferential Moment	Mc	5160.48	kgf-m.
Longitudinal Moment	Ml	5160.48	kgf-m.
Torsional Moment	Mt	6192.58	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (15.415) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Dv (.543) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.292) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 686.725 + 2 \cdot 1.65 \cdot \sqrt{(591.0(82.0 - 0.0))}$$

$$= 1413.190 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1182.000	mm
Nozzle Outside Diameter	(d) =	686.725	mm
Vessel Thickness used	(T) =	82.000	mm
Nozzle Thickness used	(t) =	158.500	mm

T / t = 0.517
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.206$$

Nr/P = 0.067
 Mr/P = 0.131
 M0/P = 0.039
 N0/P = 0.094
 MrD/Mc = 0.278
 NrDL/Mc = 0.102
 M0d/Mc = 0.086
 N0DL/Mc = 0.073
 MrD/Ml = 0.107
 NrDL/Ml = 0.061
 M0D/Ml = 0.034
 N0DL/Ml = 0.117

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	1018	1181	1018	1181
Outplane Stress Summary	1006	1180	1077	1162

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-71	71	71	-71
Inplane Pressure Stress	509	509	509	509
Inplane Stress Summary	491	522	649	392

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	14	14	6	6

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1006	1181	1077	1162

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-57	57	57	-57
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	509	509	509	509
Outplane Stress Summary	466	548	598	449

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-186	186	186	-186
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1018	1181	1018	1181
Inplane Stress Summary	881	1304	1277	954

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	10	10	10	10
Shear Stress Summary	6	6	14	14

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	881	1304	1277	955

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	159	355	159	355
Hoop Stress Summary	153	349	179	375

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-13	13	13	-13



Axial Pressure Stress	79	79	79	79
-----------------------	----	----	----	----

Axial Stress Summary	61	66	124	74
----------------------	----	----	-----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5

Shear Stress Summary	7	7	3	3
----------------------	---	---	---	---

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	154	349	179	375
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	159	355	159	355

Hoop Stress Summary	158	354	174	370
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-40	40	40	-40
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	79	79	79	79

Axial Stress Summary	34	93	151	47
----------------------	----	----	-----	----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	5	5	5	5

Shear Stress Summary	3	3	7	7
----------------------	---	---	---	---

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	158	354	176	370
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1018	1181	1018	1181	1018	1181	1018	1181
Circ. Pl (SUS)	-7	-7	19	19	-7	-7	15	15
Circ. Q (SUS)	-6	6	38	-38	-130	130	242	-242
Long. Pm (SUS)	509	509	509	509	509	509	509	509
Long. Pl (SUS)	-2	-2	10	10	-2	-2	14	14
Long. Q (SUS)	-15	15	127	-127	-41	41	73	-73
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	1018	1181	1018	1181	1018	1181	1018	1181
Pm+Pl (SUS)	1011	1174	1037	1200	1011	1174	1033	1196
Pm+Pl+Q (Total)	1005	1180	1075	1162	881	1304	1275	954

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1181.00	1325.61	Passed
Pm+Pl (SUS)	1200.02	1988.41	Passed
Pm+Pl+Q (TOTAL)	1304.05	3976.82	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	159	355	159	355	159	355	159	355
Circ. Pl (SUS)	-7	-7	19	19	-2	-2	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	79	79	79	79	79	79	79	79
Long. Pl (SUS)	-16	-16	18	18	-16	-16	18	18
Long. Q (SUS)	-2	2	24	-24	-29	29	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	159.0	355.0	159.0	355.0	159.0	355.0	159.0	355.0
Pm+Pl (SUS)	152.0	348.0	178.0	374.0	157.0	353.0	173.1	369.0
Pm+Pl+Q (Total)	152.5	348.2	178.2	374.0	157.1	353.0	174.8	369.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	355.00	1325.61	Passed
Pm+Pl (SUS)	374.01	1988.41	Passed
Pm+Pl+Q (TOTAL)	374.03	3976.82	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : 401-E-31002-AK
Nozzle Calcs.: N1-Tube inlet

Nozl: 9 1:29pm

Page 114 of 162
Jul 4, 2026



HEXAGON

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: Noz N2 Fr30

From: 30

Pressure for Reinforcement Calculations	P	163.161	kgf/cm ²
Temperature for Internal Pressure	Temp	450	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	450	°C

Parent Material		SA-336 F11	
Parent Allowable Stress at Temperature	Sv	1325.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	2235.0000	mm
Shell Finished (Minimum) Thickness	t	82.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		165.82	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1325.61	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		500.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
-----------------	--	--------	--

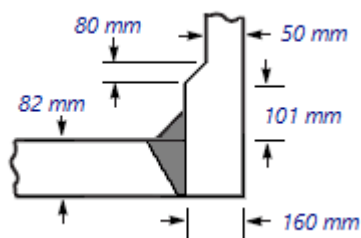
Hub Height of Integral Nozzle	h	101.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		2500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: Noz N2 Fr30

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 408.026 mm.
Actual Thickness Used in Calculation 49.987 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $163 \cdot 550 / (1326 \cdot 1 - 0.6 \cdot 163)$
= 73.0943 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $163 \cdot 206 / (1326 \cdot 1 - 0.6 \cdot 163)$
= 27.3124 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.2365 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 110/0.5773
= 190.5478 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

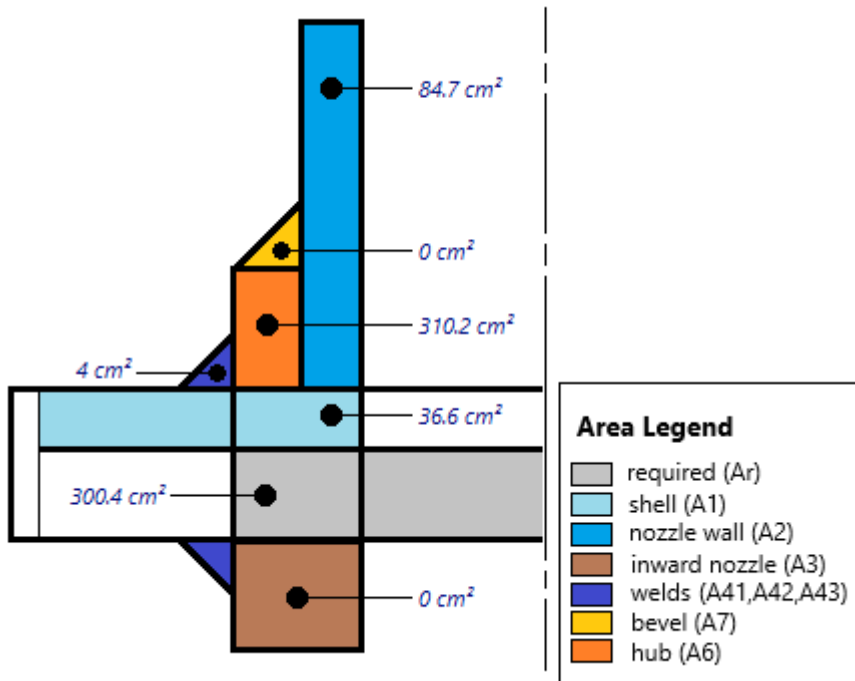
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	822.0513	mm
Parallel to Vessel Wall, opening length	d	411.0256	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	205.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1326/1326)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1326/1326)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	300.436	11.783	NA
Area in Shell	A1	36.605	313.475	NA
Area in Nozzle Wall	A2	84.699	189.003	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.000	4.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	310.236	310.236	NA
TOTAL AREA AVAILABLE	Atot	435.540	816.714	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (411 \cdot 73.1 \cdot 1 + 2 \cdot 158 \cdot 73.1 \cdot 1 \cdot (1 - 1))$
 $= 300.436 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 411(1 \cdot 82 - 1 \cdot 73.1) - 2 \cdot 158$
 $(1 \cdot 82 - 1 \cdot 73.1) \cdot (1 - 1)$
 $= 36.605 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho)(tn - trn)fr2$
 $= (2 \cdot 200)(48.5 - 27.3)1$
 $= 84.699 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 20^2 \cdot 1 + (0)^2 \cdot 1$
 $= 4.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(200, 200, 101)) \cdot (160 - 50) \cdot 1 \\ &= 310.236 \text{ cm}^2 \text{ Includes Bevel Area of } 88.010 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 28.8124 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 4.0000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 73.0943 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 73.0943 \text{ mm} \\ \text{Wall Thickness, shell/head, external pressure} & \quad t_{rb2} = 0.4378 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.0000 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 9.8312 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[9.83, \max(73.1, 4)] \\ &= 9.8312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(28.8, 9.83) \\ &= 28.8124 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 49.987 = 43.739 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 424.2,	Allowable	: 1325.6 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2991.8 kgf/cm ²	Passed
Occasional	: 309.3,	Allowable	: 1763.1 kgf/cm ²	Passed
Shear	: 90.3,	Allowable	: 927.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 43.7$, $t_r = 25.6$, $c = 1.5 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.61$, Temp. Reduction = $22 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	11 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 82$, $t_r = 68.5$, $c = 0 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = $9 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	18 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $163.16/433.49 = 0.376$

Weld Size Calculations, Description: Noz N2 Fr30

Intermediate Calc. for nozzle/shell welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (300 - 36.6 + 2 • 158 • 1 •
(1 • 82 - 73.1))1326)
= 387159.91 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (395 + 4 - 0 • 1) • 1326
= 528830.62 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (395 + 0 + 4 + (260)) • 1326
= 873408.38 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (395 + 0 + 4 + 0 + (260)) • 1326
= 873408.38 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 728 • 20 • 0.49 • 1326
= 148562. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 285) • (160 - 1.5) • 0.7 • 1326
= 1315754. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 728 • (24 - 0) • 0.74 • 1326
= 269230. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (148562 + 1315753) = 1464315 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (148562 + 0 + 269230 + 0) = 417792 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (148562 + 269230 + 0) = 417792 kgf

Summary of Failure Path Calculations:

Path 1-1 = 1464315 kgf, must exceed W = 387159 kgf	or W1 = 528830 kgf
Path 2-2 = 417791 kgf, must exceed W = 387159 kgf	or W2 = 873408 kgf
Path 3-3 = 417791 kgf, must exceed W = 387159 kgf	or W3 = 873408 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 173 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 137.6959 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 419.6959 mm

Input Echo, WRC297 Item 1, Description: Noz N2 Fr30 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	82.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-336 F11	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1325.61	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	408.026	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1325.61	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-5353.79	kgf
Circumferential Shear	Vc	4079.06	kgf
Longitudinal Shear	Vl	4079.06	kgf
Circumferential Moment	Mc	6370.96	kgf-m.
Longitudinal Moment	Ml	6370.96	kgf-m.
Torsional Moment	Mt	9250.64	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (15.415) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Dv (.576) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.550) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 728.026 + 2 \cdot 1.65 \cdot \sqrt{(591.0(82.0 - 0.0))}$$

$$= 1454.490 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1182.000	mm
Nozzle Outside Diameter	(d) =	728.026	mm
Vessel Thickness used	(T) =	82.000	mm
Nozzle Thickness used	(t) =	158.500	mm

T / t = 0.517
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.338$$

Nr/P = 0.065
 Mr/P = 0.126
 M0/P = 0.038
 N0/P = 0.093
 MrD/Mc = 0.275
 NrDL/Mc = 0.103
 M0d/Mc = 0.085
 N0DL/Mc = 0.076
 MrD/Ml = 0.105
 NrDL/Ml = 0.058
 M0D/Ml = 0.032
 N0DL/Ml = 0.114

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	1018	1181	1018	1181
Outplane Stress Summary	1004	1180	1083	1160

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-81	81	81	-81
Inplane Pressure Stress	509	509	509	509
Inplane Stress Summary	484	528	663	379

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	13	13	13	13
Shear Stress Summary	17	17	9	9

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1004	1181	1083	1160

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-66	66	66	-66
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	509	509	509	509
Outplane Stress Summary	458	554	610	441

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-214	214	214	-214
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1018	1181	1018	1181
Inplane Stress Summary	855	1328	1311	925

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	13	13	13	13
Shear Stress Summary	9	9	17	17

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	855	1328	1312	925

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	183	378	183	378
Hoop Stress Summary	175	371	205	401

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-15	15	15	-15



Axial Pressure Stress	91	91	91	91
-----------------------	----	----	----	----

Axial Stress Summary	70	78	139	84
----------------------	----	----	-----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	9	9	4	4

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	176	371	205	401

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	183	378	183	378
Hoop Stress Summary	180	376	200	396

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-46	46	46	-46
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	91	91	91	91
Axial Stress Summary	39	109	170	53

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	9	9

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	180	376	202	396

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1018	1181	1018	1181	1018	1181	1018	1181
Circ. Pl (SUS)	-7	-7	21	21	-8	-8	18	18
Circ. Q (SUS)	-6	6	42	-42	-154	154	274	-274
Long. Pm (SUS)	509	509	509	509	509	509	509	509
Long. Pl (SUS)	-2	-2	12	12	-2	-2	16	16
Long. Q (SUS)	-21	21	141	-141	-48	48	84	-84
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	1018	1181	1018	1181	1018	1181	1018	1181
Pm+Pl (SUS)	1011	1174	1039	1202	1010	1173	1036	1199
Pm+Pl+Q (Total)	1005	1180	1081	1160	856	1327	1310	925

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1181.00	1325.61	Passed
Pm+Pl (SUS)	1202.02	1988.41	Passed
Pm+Pl+Q (TOTAL)	1327.10	3976.82	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	183	378	183	378	183	378	183	378
Circ. Pl (SUS)	-7	-7	21	21	-2	-2	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	91	91	91	91	91	91	91	91
Long. Pl (SUS)	-17	-17	19	19	-17	-17	19	19
Long. Q (SUS)	-4	4	26	-26	-35	35	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	183.0	378.0	183.0	378.0	183.0	378.0	183.0	378.0
Pm+Pl (SUS)	176.0	371.0	204.0	399.0	181.0	376.0	199.0	394.0
Pm+Pl+Q (Total)	176.6	371.2	204.2	399.1	181.1	376.1	200.9	394.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	378.00	1325.61	Passed
Pm+Pl (SUS)	399.01	1988.41	Passed
Pm+Pl+Q (TOTAL)	399.05	3976.82	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : 401-E-31002-AK
Nozzle Calcs.: Noz N2 Fr30

Nozl: 10 1:29pm

Page 125 of 162
Jul 4, 2026



HEXAGON

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N3-Shell Inlet

From: 60

Pressure for Reinforcement Calculations	P	50.077	kgf/cm ²
Temperature for Internal Pressure	Temp	450	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	450	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1458.19	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	6236.4663	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		353.53	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	988.09	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		222.2000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	25.4000	mm

Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	

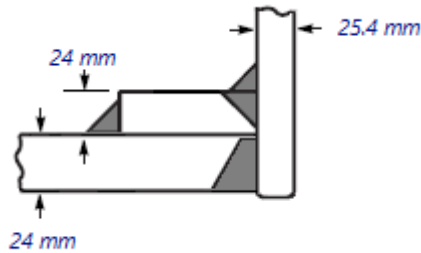
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1458.19	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	373.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		900	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	222.200 mm.
Actual Thickness Used in Calculation	25.400 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.1 \cdot 552 / (1458 \cdot 1 - 0.6 \cdot 50.1)$
 $= 19.3381 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.1 \cdot 113 / (988 \cdot 1 - 0.6 \cdot 50.1)$
 $= 5.8856 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0175 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	450.4000 mm
Parallel to Vessel Wall, opening length	d	225.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		56.2500 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 988 / 1458)$
 $= 0.678$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 988 / 1458)$
 $= 0.678$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 1458 / 1458)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.68, 1)$
 $= 0.678$

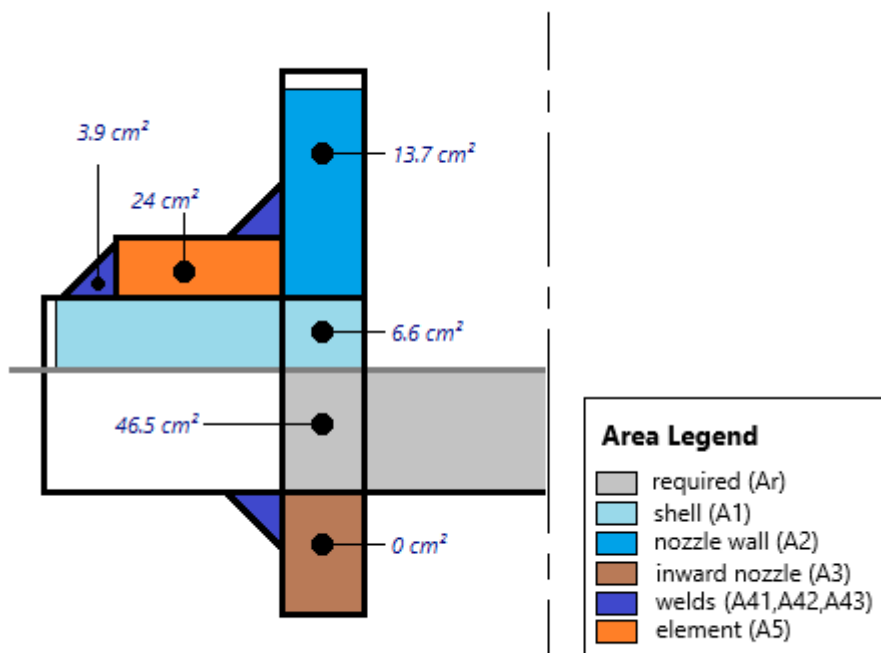


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	46.529	10.136	NA
Area in Shell	A1	6.633	29.527	NA
Area in Nozzle Wall	A2	13.733	17.444	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.918	3.918	NA
Area in Element	A5	24.000	24.000	NA
TOTAL AREA AVAILABLE	Atot	48.284	74.888	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 365.6898 24.0000 mm
 Based on given Pad Diameter: 373.0001 22.2456 mm
 Based on Shell or Nozzle Thickness: 365.6898 24.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (225 \cdot 19.3 \cdot 1 + 2 \cdot 23.9 \cdot 19.3 \cdot 1 \cdot (1 - 0.68))$
 $= 46.529 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 225(1 \cdot 22.5 - 1 \cdot 19.3) - 2 \cdot 23.9$
 $(1 \cdot 22.5 - 1 \cdot 19.3) \cdot (1 - 0.68)$
 $= 6.633 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 56.3) \cdot (23.9 - 5.89) \cdot 0.68$



= 13.733 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10^2 \cdot 0.68 + (0)^2 \cdot 0.68 + 18^2 \cdot 1$
= 3.918 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(373 - 273) \cdot 24 \cdot 1$
= 24.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.3856 mm
Wall Thickness per UG16(b),	tr16b = 4.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 20.8381 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 20.8381 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.8990 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.6, max(20.8, 4)]
= 9.6026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(7.39, 9.6)
= 9.6026 mm

Available Nozzle Neck Thickness = 25.4000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 250.1,	Allowable	: 988.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2489.0 kgf/cm ²	Passed
Occasional	: 106.6,	Allowable	: 1314.2 kgf/cm ²	Passed
Shear	: 104.8,	Allowable	: 691.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 25.4, tr = 4.81, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 24, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.83$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 24, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.83$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	18 °C
--	-------



Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 24, tr = 18.7, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.83$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 24, tr = 18.7, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.83$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Gov. MDMT of the Nozzle : 8 °C
 Gov. MDMT of the Reinforcement Pad : 8 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 8 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $50.08 / 158.16 = 0.317$

Weld Size Calculations, Description: N3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (46.5 - 6.63 + 2 \cdot 23.9 \cdot 0.68 \cdot (1 \cdot 22.5 - 19.3)) \cdot 1458)$
 $= 59669.18 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (13.7 + 24 + 3.92 - 0 \cdot 0.68) \cdot 1458$
 $= 60734.08 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (13.7 + 0 + 0.68 + (7.29)) \cdot 1458$
 $= 31639.85 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (13.7 + 0 + 3.92 + 24 + (7.29)) \cdot 1458$
 $= 71361.00 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 988$

= 20762. kgf

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 373 \cdot 18 \cdot 0.49 \cdot 1458$$

$$= 75355. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 125) \cdot (25.4 - 1.5) \cdot 0.7 \cdot 988$$

$$= 64682. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 273 \cdot 24 \cdot 0.74 \cdot 1458$$

$$= 111055. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 273 \cdot (24 - 1.5) \cdot 0.74 \cdot 1458$$

$$= 104115. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (75355 + 64682) = 140037 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20762 + 111055 + 104115 + 0) = 235932 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (75355 + 104115 + 0) = 179470 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 140037 kgf, must exceed W = 59669 kgf or W1 = 60734 kgf
 Path 2-2 = 235932 kgf, must exceed W = 59669 kgf or W2 = 31639 kgf
 Path 3-3 = 179469 kgf, must exceed W = 59669 kgf or W3 = 71361 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 50.9 kgf/cm²

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 17.2076 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 341.2076 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1458.19	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	222.200	mm
Nozzle Thickness	25.4000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 988.09	kgf/cm ²



See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 373.000 mm
 Thickness of Reinforcing Pad 24.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	50.08	kgf/cm ²
Radial Load	P	-856.60	kgf
Circumferential Shear	Vc	1274.71	kgf
Longitudinal Shear	Vl	1274.71	kgf
Circumferential Moment	Mc	948.00	kgf-m.
Longitudinal Moment	Ml	1274.19	kgf-m.
Torsional Moment	Mt	1590.20	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.0 + 2 \cdot 1.65 \cdot \sqrt{(574.75(24.0 - 1.5))}$
 $= 648.271 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1149.500 mm
Nozzle Outside Diameter	(d) =	273.000 mm
Vessel Thickness used	(T) =	46.500 mm
Nozzle Thickness used	(t) =	23.900 mm
T / t	=	1.946
d / t	=	11.423

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.181$$

Nr/P	=	0.131
Mr/P	=	0.124
M0/P	=	0.066
N0/P	=	0.154
MrD/Mc	=	0.214
NrDL/Mc	=	0.183
M0d/Mc	=	0.146
N0DL/Mc	=	0.052
MrD/Ml	=	0.162
NrDL/Ml	=	0.141
M0D/Ml	=	0.111
N0DL/Ml	=	0.170

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-36	-36	36	36
Outplane Bending (ML)	-143	143	143	-143
Normal Pressure Stress	569	619	569	619
Outplane Stress Summary	411	717	771	503

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-30	-30	30	30
Inplane Bending (ML)	-209	209	209	-209
Inplane Pressure Stress	284	284	284	284
Inplane Stress Summary	79	440	559	81

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	6	6	-6	-6
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	29	29	29	29
Shear Stress Summary	35	35	22	22

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	415	721	773	505

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-140	140	140	-140
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	284	284	284	284
Outplane Stress Summary	157	407	455	142

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-29	-29	29	29
Inplane Bending (Mc)	-205	205	205	-205
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	569	619	569	619
Inplane Stress Summary	369	772	839	419

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-6	-6	6	6



Torsional Shear (Mt)	29	29	29	29
----------------------	----	----	----	----

Shear Stress Summary	22	22	35	35
----------------------	----	----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	371	773	842	423

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-36	-36	36	36
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	255	316	255	316
Hoop Stress Summary	225	285	298	358

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-118	-118	118	118
Axial Bending (ML)	-448	448	448	-448
Axial Pressure Stress	127	127	127	127
Axial Stress Summary	-382	410	752	-250

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	12	12	-12	-12
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	56	56	56	56
Shear Stress Summary	69	69	44	44

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	623	441	756	615

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	255	316	255	316



Hoop Stress Summary 253 313 270 330

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	-88	-88	88	88
Axial Bending (Mc)	-445	445	445	-445
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	127	127	127	127
Axial Stress Summary	-349	437	718	-277

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-12	-12	12	12
Shear due to Torsion	56	56	56	56
Shear Stress Summary	44	44	69	69

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	609	451	729	623

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 273.0 + 2 \cdot 1.65 \cdot \sqrt{562.75(24.0 - 1.5)}$
 $= 644.332 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1125.500 mm
 Nozzle Outside Diameter (d) = 373.000 mm
 Vessel Thickness used (T) = 22.500 mm
 Nozzle Thickness used (t) = 73.900 mm
 T / t = 0.304
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.344$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.065
 Mr/P = 0.128
 M0/P = 0.038
 N0/P = 0.079
 MrD/Mc = 0.280
 NrDL/Mc = 0.094
 M0d/Mc = 0.085
 N0DL/Mc = 0.081
 MrD/Ml = 0.101
 NrDL/Ml = 0.072
 M0D/Ml = 0.030
 N0DL/Ml = 0.100

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom

longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	39	-39	39	-39
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-67	-67	67	67
Outplane Bending (ML)	-122	122	122	-122
Normal Pressure Stress	1202	1252	1202	1252
Outplane Stress Summary	1064	1282	1445	1172

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	129	-129	129	-129
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-48	-48	48	48
Inplane Bending (ML)	-410	410	410	-410
Inplane Pressure Stress	601	601	601	601
Inplane Stress Summary	283	843	1200	120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	32	32	32	32
Shear Stress Summary	41	41	22	22

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1067	1286	1447	1172

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	39	-39	39	-39
Outplane Membrane (Mc)	-40	-40	40	40
Outplane Bending (Mc)	-256	256	256	-256
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	601	601	601	601
Outplane Stress Summary	356	792	950	359

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	129	-129	129	-129
Inplane Membrane (Mc)	-47	-47	47	47
Inplane Bending (Mc)	-842	842	842	-842
Inplane Membrane (ML)	0	0	0	0

Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1202	1252	1202	1252
Inplane Stress Summary	453	1928	2233	339

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	32	32	32	32
Shear Stress Summary	22	22	41	41

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	458	1929	2234	392

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	569	619	569	619	569	619	569	619
Circ. Pl (SUS)	-30	-30	42	42	-24	-24	34	34
Circ. Q (SUS)	-128	128	158	-158	-176	176	234	-234
Long. Pm (SUS)	284	284	284	284	284	284	284	284
Long. Pl (SUS)	-25	-25	35	35	-2	-2	14	14
Long. Q (SUS)	-180	180	238	-238	-125	125	155	-155
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	29	29	29	29	29	29	29	29
Pm (SUS)	569.0	619.0	569.0	619.0	569.0	619.0	569.0	619.0
Pm+Pl (SUS)	539.1	589.1	611.1	661.1	545.1	595.1	603.1	653.1
Pm+Pl+Q (Total)	414.6	721.3	771.5	504.2	371.5	772.4	840.2	423.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	619.00	1458.19	Passed
Pm+Pl (SUS)	661.11	2187.29	Passed
Pm+Pl+Q (TOTAL)	840.16	4374.58	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1202	1252	1202	1252	1202	1252	1202	1252

Circ. P1 (SUS)	-54	-54	80	80	-37	-37	57	57
Circ. Q (SUS)	-83	83	161	-161	-713	713	971	-971
Long. Pm (SUS)	601	601	601	601	601	601	601	601
Long. P1 (SUS)	-38	-38	58	58	-27	-27	53	53
Long. Q (SUS)	-281	281	539	-539	-217	217	295	-295
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	32	32	32	32	32	32	32	32
Pm (SUS)	1202	1252	1202	1252	1202	1252	1202	1252
Pm+P1 (SUS)	1148	1198	1282	1332	1165	1215	1259	1309
Pm+P1+Q (Total)	1067	1284	1445	1171	457	1928	2231	390

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1252.00	1458.19	Passed
Pm+P1 (SUS)	1332.12	2187.29	Passed
Pm+P1+Q (TOTAL)	2231.31	4374.58	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	255	316	255	316	255	316	255	316
Circ. P1 (SUS)	-30	-30	42	42	-2	-2	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	127	127	127	127	127	127	127	127
Long. P1 (SUS)	-114	-114	122	122	-84	-84	92	92
Long. Q (SUS)	-396	396	500	-500	-393	393	497	-497
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	56	56	56	56	56	56	56	56
Pm (SUS)	255.0	316.0	255.0	316.0	255.0	316.0	255.0	316.0
Pm+P1 (SUS)	225.7	286.5	299.8	359.3	253.7	314.5	271.7	331.3
Pm+P1+Q (Total)	623.0	439.2	753.2	615.3	609.4	450.2	726.1	623.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	316.00	988.09	Passed
Pm+P1 (SUS)	359.31	1482.14	Passed
Pm+P1+Q (TOTAL)	753.24	2964.28	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N4-Shell outlet

From: 60

Pressure for Reinforcement Calculations	P	49.967	kgf/cm ²
Temperature for Internal Pressure	Temp	450	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	450	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1458.19	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	6236.4663	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		793.53	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	988.09	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		257.1600	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	33.3200	mm

Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	

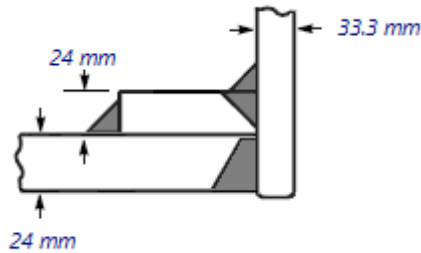
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1458.19	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	423.8000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		900	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 257.160 mm.
 Actual Thickness Used in Calculation 33.320 mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50 \cdot 552 / (1458 \cdot 1 - 0.6 \cdot 50)$
 $= 19.2947 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50 \cdot 130 / (988 \cdot 1 - 0.6 \cdot 50)$
 $= 6.7839 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1253 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	520.3200 mm
Parallel to Vessel Wall, opening length	d	260.1600 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		56.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 988 / 1458)$
 $= 0.678$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 988 / 1458)$
 $= 0.678$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1458 / 1458)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.68, 1)$
 $= 0.678$

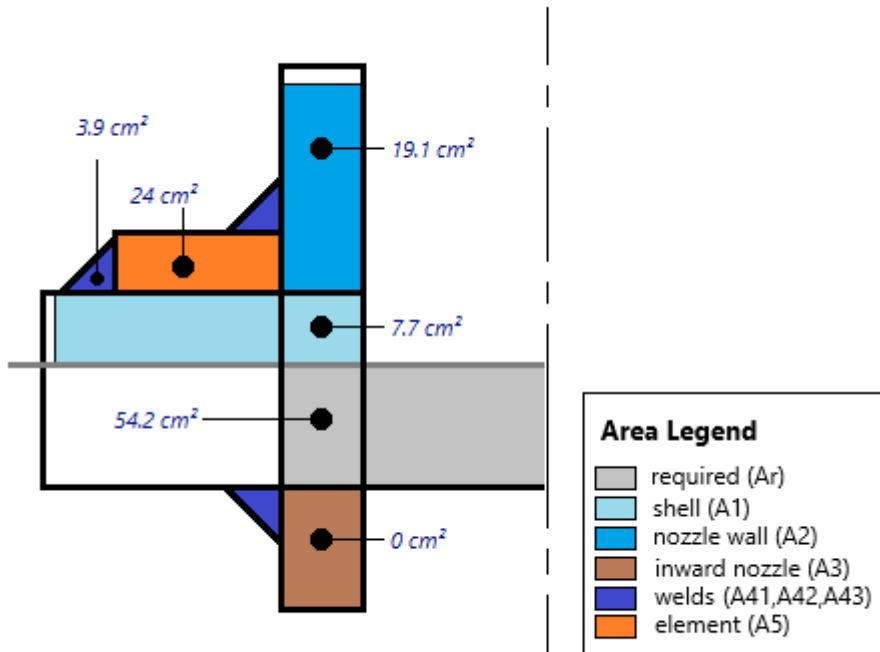


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	54.156	11.824	NA
Area in Shell	A1	7.681	33.728	NA
Area in Nozzle Wall	A2	19.085	23.399	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.918	3.918	NA
Area in Element	A5	24.000	24.000	NA
TOTAL AREA AVAILABLE	Atot	54.684	85.045	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 421.5974 24.0000 mm
 Based on given Pad Diameter: 423.8000 23.4714 mm
 Based on Shell or Nozzle Thickness: 421.5974 24.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (260 \cdot 19.3 \cdot 1 + 2 \cdot 31.8 \cdot 19.3 \cdot 1 \cdot (1 - 0.68))$
 $= 54.156 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 260(1 \cdot 22.5 - 1 \cdot 19.3) - 2 \cdot 31.8$
 $(1 \cdot 22.5 - 1 \cdot 19.3) \cdot (1 - 0.68)$
 $= 7.681 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 56.3) \cdot (31.8 - 6.78) \cdot 0.68$



= 19.085 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10^2 \cdot 0.68 + (0)^2 \cdot 0.68 + 18^2 \cdot 1$
= 3.918 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(424 - 324) \cdot 24 \cdot 1$
= 24.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 8.2839 mm
Wall Thickness per UG16(b),	tr16b = 4.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 20.7947 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 20.7947 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.8990 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.83, max(20.8, 4)]
= 9.8312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(8.28, 9.83)
= 9.8312 mm

Available Nozzle Neck Thickness = 33.3200 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 205.5,	Allowable	: 988.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2533.7 kgf/cm ²	Passed
Occasional	: 91.0,	Allowable	: 1314.2 kgf/cm ²	Passed
Shear	: 84.4,	Allowable	: 691.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 33.3, tr = 5.54, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.17$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 24, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.83$, Temp. Reduction = 10 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 24, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.83$, Temp. Reduction = 10 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	18 °C
--	-------

Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 24, tr = 18.6, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.83$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 24, tr = 18.6, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.83$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

Gov. MDMT of the Nozzle : 8 °C
 Gov. MDMT of the Reinforcement Pad : 8 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 8 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $49.97/158.16 = 0.316$

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (54.2 - 7.68 + 2 \cdot 31.8 \cdot 0.68 \cdot (1 \cdot 22.5 - 19.3)) \cdot 1458)$
 $= 69784.17 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (19.1 + 24 + 3.92 - 0 \cdot 0.68) \cdot 1458$
 $= 68539.43 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (19.1 + 0 + 0.68 + (9.7)) \cdot 1458$
 $= 42966.77 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (19.1 + 0 + 3.92 + 24 + (9.7)) \cdot 1458$
 $= 82687.91 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 988$

= 24626. kgf

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 424 \cdot 18 \cdot 0.49 \cdot 1458$$

$$= 85618. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 146) \cdot (33.3 - 1.5) \cdot 0.7 \cdot 988$$

$$= 100941. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 324 \cdot 24 \cdot 0.74 \cdot 1458$$

$$= 131721. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 324 \cdot (24 - 1.5) \cdot 0.74 \cdot 1458$$

$$= 123488. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (85618 + 100941) = 186559 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (24626 + 131721 + 123488 + 0) = 279835 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (85618 + 123488 + 0) = 209106 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 186558 kgf, must exceed W = 69784 kgf or W1 = 68539 kgf
 Path 2-2 = 279834 kgf, must exceed W = 69784 kgf or W2 = 42966 kgf
 Path 3-3 = 209106 kgf, must exceed W = 69784 kgf or W3 = 82687 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 50.2 kgf/cm²

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 24.3686 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.3686 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1458.19	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	257.160	mm
Nozzle Thickness	33.3200	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 988.09	kgf/cm ²



See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 423.800 mm
 Thickness of Reinforcing Pad 24.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	49.97	kgf/cm ²
Radial Load	P	-1029.96	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1529.65	kgf
Circumferential Moment	Mc	1376.13	kgf-m.
Longitudinal Moment	Ml	1834.84	kgf-m.
Torsional Moment	Mt	2293.54	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.718) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 323.8 + 2 \cdot 1.65 \cdot \sqrt{574.75(24.0 - 1.5)}$$

$$= 699.071 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1149.500 mm
Nozzle Outside Diameter	(d) =	323.800 mm
Vessel Thickness used	(T) =	46.500 mm
Nozzle Thickness used	(t) =	31.820 mm
T / t	=	1.461
d / t	=	10.176

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.401$$

Nr/P	=	0.121
Mr/P	=	0.137
M0/P	=	0.057
N0/P	=	0.132
MrD/Mc	=	0.240
NrDL/Mc	=	0.166
M0d/Mc	=	0.122
N0DL/Mc	=	0.050
MrD/Ml	=	0.159
NrDL/Ml	=	0.119
M0D/Ml	=	0.084
N0DL/Ml	=	0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-38	-38	38	38
Outplane Bending (ML)	-131	131	131	-131
Normal Pressure Stress	568	618	568	618

Outplane Stress Summary 420 701 761 515

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-31	-31	31	31
Inplane Bending (ML)	-249	249	249	-249
Inplane Pressure Stress	284	284	284	284

Inplane Stress Summary 48 469 609 32

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	6	6	-6	-6
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	29	29	29	29

Shear Stress Summary 36 36 23 23

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 424 706 765 516

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-144	144	144	-144
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	284	284	284	284

Outplane Stress Summary 152 408 461 139

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	-32	-32	32	32
Inplane Bending (Mc)	-283	283	283	-283
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	568	618	568	618

Inplane Stress Summary 297 835 929 334

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-6	-6	6	6
Torsional Shear (Mt)	29	29	29	29

Shear Stress Summary	23	23	36	36
----------------------	----	----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	301	837	932	340
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-38	-38	38	38
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	218	278	218	278

Hoop Stress Summary	185	245	263	323
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	46	-46	46	-46
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-94	-94	94	94
Axial Bending (ML)	-332	332	332	-332
Axial Pressure Stress	109	109	109	109

Axial Stress Summary	-267	304	586	-172
----------------------	------	-----	-----	------

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	9	9	-9	-9
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	43	43	43	43

Shear Stress Summary	53	53	34	34
----------------------	----	----	----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	465	335	590	500
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	218	278	218	278
----------------------	-----	-----	-----	-----

Hoop Stress Summary	214	274	234	294
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	46	-46	46	-46
Axial Membrane (Mc)	-70	-70	70	70
Axial Bending (Mc)	-396	396	396	-396
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	109	109	109	109

Axial Stress Summary	-307	391	626	-259
----------------------	------	-----	-----	------

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-9	-9	9	9
Shear due to Torsion	43	43	43	43

Shear Stress Summary	34	34	53	53
----------------------	----	----	----	----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	526	400	633	564
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 323.8 + 2 \cdot 1.65 \cdot \sqrt{562.75(24.0 - 1.5)}$
 $= 695.132 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1125.500 mm
Nozzle Outside Diameter	(d) =	423.800 mm
Vessel Thickness used	(T) =	22.500 mm
Nozzle Thickness used	(t) =	81.820 mm
T / t	=	0.275
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.663$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.116
M0/P	=	0.035
N0/P	=	0.072
MrD/Mc	=	0.274
NrDL/Mc	=	0.094
M0d/Mc	=	0.082
N0DL/Mc	=	0.084
MrD/Ml	=	0.093
NrDL/Ml	=	0.067
M0D/Ml	=	0.027
N0DL/Ml	=	0.090

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	43	-43	43	-43
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-76	-76	76	76
Outplane Bending (ML)	-137	137	137	-137
Normal Pressure Stress	1200	1250	1200	1250
Outplane Stress Summary	1043	1282	1472	1160

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	141	-141	141	-141
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-57	-57	57	57
Inplane Bending (ML)	-477	477	477	-477
Inplane Pressure Stress	600	600	600	600
Inplane Stress Summary	219	890	1289	50

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	36	36	36	36
Shear Stress Summary	46	46	25	25

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1046	1288	1475	1161

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	43	-43	43	-43
Outplane Membrane (Mc)	-53	-53	53	53
Outplane Bending (Mc)	-317	317	317	-317
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	600	600	600	600
Outplane Stress Summary	286	835	1028	308

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	141	-141	141	-141
Inplane Membrane (Mc)	-59	-59	59	59



Inplane Bending (Mc)	-1053	1053	1053	-1053
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1200	1250	1200	1250

Inplane Stress Summary	241	2114	2467	127
------------------------	-----	------	------	-----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	36	36	36	36

Shear Stress Summary	25	25	46	46
----------------------	----	----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	298	2114	2469	319

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	568	618	568	618	568	618	568	618
Circ. Pl (SUS)	-32	-32	44	44	-27	-27	37	37
Circ. Q (SUS)	-115	115	147	-147	-244	244	322	-322
Long. Pm (SUS)	284	284	284	284	284	284	284	284
Long. Pl (SUS)	-26	-26	36	36	-3	-3	15	15
Long. Q (SUS)	-210	210	288	-288	-128	128	160	-160
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	29	29	29	29	29	29	29	29
Pm (SUS)	568.0	618.0	568.0	618.0	568.0	618.0	568.0	618.0
Pm+Pl (SUS)	536.1	586.1	612.1	662.1	541.1	591.1	605.1	655.1
Pm+Pl+Q (Total)	424.3	706.1	762.4	516.1	300.6	836.2	929.6	339.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	618.00	1458.19	Passed
Pm+Pl (SUS)	662.11	2187.29	Passed
Pm+Pl+Q (TOTAL)	929.60	4374.58	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	1200	1250	1200	1250	1200	1250	1200	1250
Circ. Pl (SUS)	-62	-62	90	90	-47	-47	71	71
Circ. Q (SUS)	-94	94	180	-180	-912	912	1194	-1194
Long. Pm (SUS)	600	600	600	600	600	600	600	600
Long. Pl (SUS)	-45	-45	69	69	-39	-39	67	67
Long. Q (SUS)	-336	336	618	-618	-274	274	360	-360
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	36	36	36	36	36	36	36	36
Pm (SUS)	1200	1250	1200	1250	1200	1250	1200	1250
Pm+Pl (SUS)	1138	1188	1290	1340	1153	1203	1271	1321
Pm+Pl+Q (Total)	1046	1287	1473	1160	298	2115	2466	318

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1250.00	1458.19	Passed
Pm+Pl (SUS)	1340.15	2187.29	Passed
Pm+Pl+Q (TOTAL)	2466.47	4374.58	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	218	278	218	278	218	278	218	278
Circ. Pl (SUS)	-32	-32	44	44	-3	-3	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	109	109	109	109	109	109	109	109
Long. Pl (SUS)	-91	-91	97	97	-67	-67	73	73
Long. Q (SUS)	-286	286	378	-378	-350	350	442	-442
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	43	43	43	43	43	43	43	43
Pm (SUS)	218.0	278.0	218.0	278.0	218.0	278.0	218.0	278.0
Pm+Pl (SUS)	186.5	246.4	263.4	322.7	215.5	275.3	234.5	293.7
Pm+Pl+Q (Total)	465.8	334.5	587.6	498.7	527.4	401.2	630.8	562.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	278.00	988.09	Passed
Pm+Pl (SUS)	322.69	1482.14	Passed
Pm+Pl+Q (TOTAL)	630.80	2964.28	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	49.97	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	49.97	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05	kgf/cm ²
Shell Thickness	ts	24.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1100.000	mm
Shell Temperature for Internal Pressure	Ts	450.00	°C
Shell Material		SA-387 11	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789	
Shell Allowable Stress at Temperature	Ss	1458.19	kgf/cm ²
Shell Allowable Stress at Ambient		1509.10	kgf/cm ²

Channel Data:

Channel Description: Shell Flg

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	163.16	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	82.0000	mm
Channel Internal Corrosion Allowance	cac	0.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1100.000	mm
Channel Design Temperature	TEMPC	470.00	°C
Channel Flange Matl		SA-336 F11	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11572	
Channel Allowable Stress at Temperature	Sc	1122.03	kgf/cm ²
Channel Allowable Stress at Ambient		1407.13	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	860	
Tube Wall Thickness	et	2.1000	mm
Tube Outside Diameter	D	25.0000	mm
Total Straight Tube Length	Lt	5800.00	mm
Straight Tube Length (bet. inner tubsht faces) L		5520.00	mm
Design Temperature of the Tubes		450.00	°C
Tube Material		SB-167	
Tube Material UNS Number		N06690	
Is this a Welded Tube		No	
Tube Material Specification used	Smls. pipe & tube		
Tube Allowable Stress at Temperature		1580.49	kgf/cm ²
Tube Allowable Stress At Ambient		1641.66	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1937.36	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	32.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.1000	mm
Groove Weld Leg	ag	2.1000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		UW-20.3	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	520.365	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm

Tubesheet Data:



Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature T 470.00 °C
 Tubesheet Material SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K21590
 Tubesheet Allowable Stress at Temperature S 1211.58 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1509.10 kgf/cm²
 Thickness of Tubesheet h 280.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 13.0000 mm
 Tubesheet Outside Diameter A 1570.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No
 Area of the Untubed Lanes AL 0.0 cm²
 Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Shell Side Min. Design Pressure Psd,min 1.0546 kgf/cm²

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Shell
 Flange Outside Diameter A 1570.000 mm
 Flange Inside Diameter B 1100.000 mm
 Flange Face Outside Diameter Fod 1146.000 mm
 Flange Face Inside Diameter Fid 1100.000 mm
 Gasket Outside Diameter Go 1135.000 mm
 Gasket Inside Diameter Gi 1103.000 mm
 Small end Hub thk. g0 24.0000 mm
 Large end Hub thk. g1 34.0000 mm
 Gasket Factor, m 2.00
 Gasket Design Seating Stress y 175.77 kgf/cm²
 Flange Facing Sketch Code Sketch 1a
 Gasket Thickness tg 3.1750 mm
 Full face Gasket Flange Option Program Selects

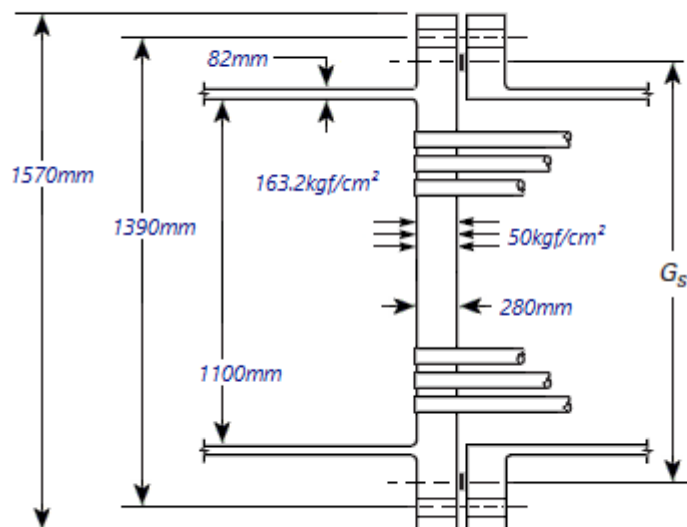
Bolting Information:

Diameter of Bolt Circle C 1390.000 mm
 Nominal Bolt Diameter dB 88.9000 mm
 Thread Series TEMA Thread Series
 Number of Bolts n 24
 Bolt Material SA-193 B16
 Bolt Allowable Stress At Temperature Sb 1384.92 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1549.89 kgf/cm²
 Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with Channel
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 248.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	450.0 °C	0.17763E+07	kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07	kgf/cm²

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	470.0 °C	0.17560E+07	kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07	kgf/cm²

Tubes - TM-4 N06690

Elastic Mod. at Tubsht. Design Temp.	470.0 °C	0.18709E+07	kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21303E+07	kgf/cm²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	470.0 °C	0.18107E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (164 \cdot (25/2 - 0)) / (1580 \cdot 1 + 0.4 \cdot 164)$$

$$= 1.2469 + 0.0000 = 1.2469 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart NFN-4

at 450.00 °C

Elastic Modulus for Material	1984246.75	kgf/cm ²
------------------------------	------------	---------------------

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1000	25.00	6040.37	11.90	50.0000	0.0077616	747.16

$$\text{EMAWP} = (4*B)/(3*(D/T)) = (4 * 747)/(3 * 11.9) = 83.7 \text{ kgf/cm}^2$$



Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.3655	25.00	6040.37	18.31	50.0000	0.0032818	686.17
$EMAWP = (4*B)/(3*(D/T)) = (4 * 686)/(3 * 18.3) = 50 \text{ kgf/cm}^2$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.3655 mm
Allowable Internal Pressure at Corroded thickness	284.65 kgf/cm ²
Required Internal Design Pressure	164.21 kgf/cm ²
Allowable External Pressure at Corroded thickness	83.68 kgf/cm ²
Required External Design Pressure	49.97 kgf/cm ²
Required Thickness due to Shell Side pressure	1.3655 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	16.000 mm
Basic Gasket Width,	$b0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	1120.750 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 520 + 25 \\ &= 1065.730 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (32 - 25) / 32 \\ &= 0.219 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= Gs / Do \text{ (Configurations d, e, f)} \\ &= 1121 / 1066 \\ &= 1.0516 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= Dc / Do \text{ (Configurations a, e, f)} \\ &= 1100 / 1066 \\ &= 1.0322 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= ltx / h \\ &= 0 / 280 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25 - 2 \cdot 2.1 (1870894 / 1810721) \cdot \\ &\quad (1572 / 1212) \cdot (0); 25 - 2 \cdot 2.1) \\ &= 25.0000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$



$$= 32 / (1 - 4 \cdot \min(0; 4 \cdot 1066 \cdot 32) / (\pi \cdot 1066^2))$$

$$= 32.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32 - 25) / 32$$

$$= 0.2187$$

$$h/p = 8.750001; \mu^* = 0.218750; E^* = 321950 \text{ kgf/cm}^2$$

$$E^*/E = 0.177802; \nu^* = 0.432718$$

Note: As h/p (8.750) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1100 + 82)82)$$

$$= 0.0058 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0058 \cdot 1756027 \cdot 82^3 / (6(1 - 0.3^2))$$

$$= 1035428544.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1100 + 82 / \cos(0)) \cdot 1035428544 / 280^3 \cdot (1 + 280 \cdot 0.0058 \cdot (1 + 1) / 2 + (280 \cdot 0.0058)^2 / 2 \cdot 1)$$

$$= 1328495.6 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1100^2 / (4 \cdot 1756027 \cdot 82) \cdot (1 - 1100 \cdot 0.3 / (2(1100 + 82)))$$

$$= 0.18075E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 1035428544 [0.001807523 (1 + 0.0058 \cdot 1 \cdot 280) - 0.000 (1/280 + 1 \cdot 0.0058/2)]$$

$$= 29720.6152 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1570 / 1066$$

$$= 1.4732$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 321950 \cdot (1328495 + 1810721 \cdot \ln(1.47))$$

$$= 3.5769$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1066^2 / 16 \cdot [(1.05 - 1) \cdot (1.05^2 + 1) \cdot 50 -$$



$$(1.03 - 1) \cdot (1.03^2 + 1) \cdot 163] \\ = -383569.8438 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]
= $M_{ts} + (\omega_c \cdot P_t) + W_* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$
= $-383570 + (4849156) + 2099599 \cdot (1390 - 1121) / (2 \cdot \pi \cdot 1066)$
= $12907971.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
= $(M_* - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$
= $((12907971) - 1066^2/32 \cdot 3.58(50 - 163)) / (1 + 3.58)$
= $5959967.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
= $M_p + D_o^2/64 \cdot (3 + \nu_*) \cdot (P_s - P_t)$
= $5959967 + 1066^2/64 \cdot (3 + 0.43) \cdot (50 - 163)$
= $-935500.1250 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
= $\max(|M_p|; |M_o|)$
= $\max(|5959967|; |-935500|)$
= $5959967.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
= $6 \cdot M / (\mu_* (h - h_g')^2)$
= $6 \cdot 5959967 / (0.22(280 - 8)^2)$
= $2209.5774 \text{ kgf/cm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
= $2 \cdot S$
= $2 \cdot 1212$
= 2423.16 kgf/cm^2

Tubesheet bending stress at final thickness [σ]:
= $6 \cdot M / (\mu_* (h - h_g')^2)$
= $6 \cdot 5822332 / (0.22(265 - 8)^2)$
= $2415.6887 \text{ kgf/cm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
= $h + C_{ats} + C_{atc}$
= $265 + 0 + 0$
= 265.1163 mm

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
= $\max(H_{reqB}; H_{reqS})$
= $\max(265; 142)$
= 265.1163 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:
= $|P_s - P_t|$
= $|50 - 163|$
= 113.191 kgf/cm^2

Shear Stress check [τ_{limit}]:
= $4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$
= $4 \cdot 0.22 \cdot 280 / 1066 \cdot \min(0.8 \cdot 1212; 0.533 \cdot 2309)$
= 222.824 kgf/cm^2

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:
= $1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc}$
= $1 / (4 \cdot 0.22) \cdot (1066 / 969) \cdot |50 - 163| + 0$



= 142.2354 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 82.000 mm
Min. Channel len. adjacent to the tubesheet is: 540.600mm

Axial Channel Membrane Stress [σ_c, m]:
= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
= $1100^2 / (4 \cdot 82 (1100 + 82) \cdot \cos(0)) \cdot 163$
= 509.216 kgf/cm²

Axial Channel Bending Stress [σ_c, b]:
= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
= $6 \cdot 22827278 / 82^2 \cdot [(0.001807523 \cdot 1 \cdot 0.0058 - 0.000 \cdot 1) \cdot 163 - 6((1 - 0.43) / 321950) \cdot 1066 / 280^3 \cdot (1 + 0.0058 \cdot 280 / 2 \cdot 1) \cdot (5959967 + 1066^2 / 32 \cdot (50 - 163))]$
= -83.144 kgf/cm²

Channel Membrane + Bending Stress [σ_c]:
= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq 1.5 \cdot S_c$)
= 592.360 kgf/cm² (Should be ≤ 1683.04)

Tube Metal Area [A_t]:
= $\pi (d_o - t) t$
= $\pi (25 - 2.1) 2.1$
= 1.511 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 2.1 (25 - 2.1) 1580$
= 2387.791 kgf

Groove Weld Strength [F_g]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not $> F_t$)
= $0.85 \cdot \pi \cdot 2.1 (25 + 0.67 \cdot 2.1) 1212$
= 1794.1523 kgf

Weld Strength Factor [f_w]:
= S / S_w where $S_w = \min(S \text{ and } S_t)$
= $1580 / 1212$
= 1.304

Design Strength Ratio [f_d]:
= 1 for full strength welds

Fillet Weld Strength [F_f]:
= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not $> F_t$)
= $0.55 \cdot \pi \cdot 2.1 (25 + 0.67 \cdot 2.1) 1212$
= 1160.9222 kgf

Minimum Length of Weld [a_r]:
= $2(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o)$
= $2(\sqrt{(0.75 \cdot 25)^2 + 1.07 \cdot 2.1 \cdot (25 - 2.1) 1.3 \cdot 1} - 0.75 \cdot 25)$
= 3.424 mm

Minimum Required Fillet Weld Leg [a_{fr}]: 1.7118 mm
Minimum Required Groove Weld Leg [a_{gr}]: 1.7118 mm

Tube-Tubesheet Jt allowable, 2388 is $\geq$ tube strength 2388 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. ($P_{sd,max} + P_{td,max}$):



Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2209.6 <=	2423.2 kgf/cm ²	Ok
Tubesheet shear stress	492.4 <=	969.3 kgf/cm ²	Ok
Stress in Channel at Tubesheet	592.4 <=	1683.0 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	265.116	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.712	2.100	Ok
Tube-Tubesheet Groove Weld Leg :	1.712	2.100	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	255.862 178.195	2005	2423 714 969	Ps+Pt D1	Ok
D2uc	183.743 ...	1367	2423 217 969	Ps+Pt D2	Ok
D3uc	265.116 ...	2210	2423 492 969	Ps+Pt D3	Ok
D4uc	34.202 ...	694	2423 5 969	Ps+Pt D4	Ok
D1c	262.476 178.195	2098	2423 753 969	Ps+Pt-c D1	Ok
D2c	187.561 ...	1407	2423 229 969	Ps+Pt-c D2	Ok
D3c	269.489 ...	2267	2423 519 969	Ps+Pt-c D3	Ok
D4c	36.907 ...	695	2423 5 969	Ps+Pt-c D4	Ok
Max:	269.4891 178.195 mm	0.936	0.777 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	2431	3879	Ok
D2uc	...	...	...	1996	3879	Ok
D3uc	...	...	...	592	1683	Ok
D4uc	...	...	...	1581	1683	Ok
D1c	...	...	...	2491	3879	Ok
D2c	...	...	...	2167	3879	Ok
D3c	...	...	...	704	1683	Ok
D4c	...	...	...	1717	3879	Ok
Max	...	...	...	0.939		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
----------------	----------	--------	-----



Tubesheet thickness :	269.489	280.000	Ok
Tubesheet thickness flanged extension :	178.195	248.000	Ok
Tube thickness :	1.366	2.100	Ok
Tube-tubesheet fillet weld leg :	1.712	2.100	Ok
Tube-tubesheet groove weld leg :	1.712	2.100	Ok

Given Shell Thickness: 24.0000 mm
 Given Channel Thickness: 82.0000 mm

Min Channel length of thk, (82.000) adj. to tubesheet: 540.600 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	188.001	1.000	137.669	1.000
Tubesheet Shear Stress	211.284	1.000	211.284	1.000
Tubesheet-Channel Junction	211.284	1.124	211.284	1.605
Tube Pressure Stress	284.650	1.000	83.681	1.000
Tubesheet Extension Stress	No Calc	No Calc	172.617	...
Minimum MAWP	188.001		83.681	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	245.221	1.000	177.081	1.000
Tubesheet Shear Stress	277.540	1.000	277.540	1.000
Tubesheet-Channel Junction	277.540	1.132	277.540	1.641
Tube Pressure Stress	295.666	1.000	102.148	1.000
Tubesheet Extension Stress	No Calc	No Calc	193.178	...
Minimum MAPnc	245.221		102.148	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Channel Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, channel thickness))
 = max(280/4, min(280, 82))
 = 82.000 mm

Thickness Ratio = 0.67, Temperature Reduction per Fig. UCS 66.1 = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 9 °C

Shell Side MDMT calculation:

Governing thickness for the shell side:
 = tubesheet thickness/4
 = 280/4
 = 70.000 mm

Thickness Ratio = 0.67, Temperature Reduction per Fig. UCS 66.1 = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 6 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1



= max(163/245, 50/102)
= 0.665

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.